Supplementary material

When Does SHAP Outperform LIME? Model- and Configuration-Dependent Results from a Paired Tabular Benchmark. Manuscript 2067810, *Computers and Informatics*, revision 1.

This file is independent of the paper: it states the whole design and every result at the level of dataset, model and seed, so that each number in the paper can be checked. Repository and archive addresses are withheld for masked review.

Design fingerprint (SHA-256 of the design configuration): `4ab8673a4a205aaf70ac514a83a21abb59c9ff951a111a6679a85dfe4d97a1fa`. Every result row carries it; the analysis refuses rows of another design.

S1. Design

Item	Setting
Split	one stratified split per dataset, test share 0.2, seed 42; shared by preprocessing, training and every seed
Training cap	20,000 rows (stratified subsample of the training split)
Seeds	Adult: (11, 22, 33, 44, 55); other datasets: (11, 22, 33)
Instances	up to 25 per stratum (TP, TN, FP, FN) of the test split; smaller strata taken whole
Stability subset	first 3 instances per stratum
Stability	T = 15 perturbed copies, Gaussian noise sigma = 0.1 on the encoded features
Faithfulness gap	top k = 5 features masked together
Sparsity threshold	0.0001
Timing subset	5 instances per stratum, 3 repetitions, 1 warm-up per explainer, one process, one thread, pinned to one core

S2. Datasets

Dataset	Source	Train rows	Test rows	Encoded features	Positive share	Role
Adult	UCI	39,032	9,758	108	0.239	primary
Breast Cancer	UCI	455	114	30	0.627	further
German Credit	UCI	800	200	61	0.700	further
Diabete	OpenM	614	154	8	0.349	numeric, lower

Dataset	Source	Train rows	Test rows	Encoded features	Positive share	Role
s	L 37					dimension
Spambase	OpenML 44	3,680	921	57	0.394	numeric, higher dimension
Phishing	OpenML 4534	8,844	2,211	68	0.557	categorical, lower dimension
Connect-4	OpenML 40668	54,045	13,512	126	0.658	categorical, higher dimension

Train rows are those available before the training cap. Connect-4: a win is the positive class; loss and draw are the negative class.

S3. Models

Model	Class	Hyperparameters
Logistic regression	`sklearn.linear_model.LogisticRegression`	solver=lbfgs, max_iter=1000, random_state=42
Random forest	`sklearn.ensemble.RandomForestClassifier`	n_estimators=100, max_depth=15, random_state=42, n_jobs=1
XGBoost	`xgboost.XGBClassifier`	n_estimators=100, max_depth=6, learning_rate=0.1, objective=binary:logistic, eval_metric=logloss, random_state=42, n_jobs=1
SVM (RBF)	`sklearn.svm.SVC`	C=1.0, kernel=rbf, probability=True, random_state=42
MLP	`sklearn.neural_network.MLPClassifier`	hidden_layer_sizes=(100), max_iter=500, random_state=42

Test accuracy of the frozen models:

Dataset	Logistic regression	Random forest	XGBoost	SVM (RBF)	MLP
Adult	0.8500	0.8580	0.8666	0.8526	0.8240
Breast Cancer	0.9825	0.9561	0.9474	0.9825	0.9649
German Credit	0.7050	0.7400	0.7400	0.7400	0.7400
Diabetes	0.7143	0.7532	0.7662	0.7532	0.7078
Spambase	0.9294	0.9425	0.9468	0.9273	0.9392
Phishing	0.9398	0.9715	0.9665	0.9593	0.9742

Dataset	Logistic regression	Random forest	XGBoost	SVM (RBF)	MLP
Connect-4	0.7979	0.8290	0.8529	0.8490	0.8676

S4. Explainers

Variant	Settings
shap	method=shap, n_background_samples=50, kernel_nsamples=500
lime	method=lime, num_samples=1000, num_features=10, kernel_width=default
lime_full	method=lime, num_samples=1000, num_features=all, kernel_width=default
anchors	method=anchors, threshold=0.95
dice	method=dice, num_counterfactuals=1

SHAP uses TreeSHAP for the random forest and XGBoost and KernelSHAP otherwise. LIME's default kernel width is 0.75 times the square root of the number of encoded features. Anchors: the attribution vector marks the anchor's features with 1. DiCE: the attribution of a feature is the absolute change in the counterfactual.

S5. Computing environment

One laptop for every run: AMD Ryzen 5 4600H (6 cores, 12 threads, 3.0 GHz base), 31.4 GB RAM, Windows 11 Home 10.0.26300, on mains power. CPython 3.11.11; scikit-learn 1.7.1, XGBoost 3.1.2, SHAP 0.50.0, LIME 0.2.0.1, NumPy 2.2.6, pandas 2.3.3, SciPy 1.16.3, alibi 0.9.6, dice-ml 0.12. The full frozen environment is part of the artifacts.

S6. Model and data fingerprints

Each run checks the model file against these hashes before it starts.

Dataset	Model	Model SHA-256 (first 16)	Data SHA-256 (first 16)
Adult	Logistic regression	`c06174920a8dfa75`	`18b42fcbbcf14068`
Adult	MLP	`329f273fa5d5688a`	`18b42fcbbcf14068`
Adult	Random forest	`a2dd7551756486ea`	`18b42fcbbcf14068`
Adult	rf_d6	`48e5d53de3368649`	`18b42fcbbcf14068`
Adult	SVM (RBF)	`2b0a0c6b236bd11d`	`18b42fcbbcf14068`
Adult	XGBoost	`6fa7994e30549e68`	`18b42fcbbcf14068`
Adult	xgb_d15	`deef8fd516743635`	`18b42fcbbcf14068`
Breast Cancer	Logistic regression	`571aaf8bb37da91f`	`765348e4109f0b13`
Breast Cancer	MLP	`9e7838f9c2a4616e`	`765348e4109f0b13`
Breast	Random forest	`ece4da48e4fcc84d`	`765348e4109f0b13`

Dataset	Model	Model SHA-256 (first 16)	Data SHA-256 (first 16)
Cancer			
Breast Cancer	SVM (RBF)	`273b476e04b1f1dd`	`765348e4109f0b13`
Breast Cancer	XGBoost	`1d03c4d9da6ab8a9`	`765348e4109f0b13`
Connect-4	Logistic regression	`873e0b72df24adfc`	`e438bafb45e36a75`
Connect-4	MLP	`d737986b44181bc3`	`e438bafb45e36a75`
Connect-4	Random forest	`213c98dad25a4fc7`	`e438bafb45e36a75`
Connect-4	SVM (RBF)	`0cbc54c363e606c5`	`e438bafb45e36a75`
Connect-4	XGBoost	`6d0ea32aa008aa18`	`e438bafb45e36a75`
Diabetes	Logistic regression	`a991171cff6e2c47`	`f9f1195b4e6a9677`
Diabetes	MLP	`6e2d11ebedfcff6c`	`f9f1195b4e6a9677`
Diabetes	Random forest	`006d37382f8dfe45`	`f9f1195b4e6a9677`
Diabetes	SVM (RBF)	`82d2e966fea86ace`	`f9f1195b4e6a9677`
Diabetes	XGBoost	`51545d5ae1130e5a`	`f9f1195b4e6a9677`
German Credit	Logistic regression	`f94432191bc002cf`	`04b09e7ae7547830`
German Credit	MLP	`4c406969284f6c6c`	`04b09e7ae7547830`
German Credit	Random forest	`fdbf9f1e3859d9c7`	`04b09e7ae7547830`
German Credit	SVM (RBF)	`0814b148809bf258`	`04b09e7ae7547830`
German Credit	XGBoost	`15ed0d95627e82d3`	`04b09e7ae7547830`
Phishing	Logistic regression	`b691431585152391`	`9d93cc662c21ea41`
Phishing	MLP	`11980a4dcd44ab8e`	`9d93cc662c21ea41`
Phishing	Random forest	`552a74bafd59aa88`	`9d93cc662c21ea41`
Phishing	SVM (RBF)	`b5ef782a7937d841`	`9d93cc662c21ea41`
Phishing	XGBoost	`c3583f6aef8d7f67`	`9d93cc662c21ea41`
Spambase	Logistic regression	`df2e6e9f26ecdc5b`	`89f80a488462e60e`
Spambase	MLP	`ec3acdf8a11764f8`	`89f80a488462e60e`

Dataset	Model	Model SHA-256 (first 16)	Data SHA-256 (first 16)
Spambase	Random forest	`b788ed367089fd13`	`89f80a488462e60e`
Spambase	SVM (RBF)	`9c3450727360d61a`	`89f80a488462e60e`
Spambase	XGBoost	`0b808e596faf5ff0`	`89f80a488462e60e`

S7. Realized instances

Quality pass: 575 of 575 jobs, 52,740 instance rows. Instances per stratum of the test split, for each dataset, model and seed (a stratum with fewer than 25 instances is taken whole), and the instances with stability computed:

Dataset	Model	Seed	TP	TN	FP	FN	Stability
Adult	Logistic regression	11	25	25	25	25	12
Adult	Logistic regression	22	25	25	25	25	12
Adult	Logistic regression	33	25	25	25	25	12
Adult	Logistic regression	44	25	25	25	25	12
Adult	Logistic regression	55	25	25	25	25	12
Adult	Random forest	11	25	25	25	25	12
Adult	Random forest	22	25	25	25	25	12
Adult	Random forest	33	25	25	25	25	12
Adult	Random forest	44	25	25	25	25	12
Adult	Random forest	55	25	25	25	25	12
Adult	XGBoost	11	25	25	25	25	12
Adult	XGBoost	22	25	25	25	25	12
Adult	XGBoost	33	25	25	25	25	12
Adult	XGBoost	44	25	25	25	25	12
Adult	XGBoost	55	25	25	25	25	12
Adult	SVM (RBF)	11	25	25	25	25	12
Adult	SVM (RBF)	22	25	25	25	25	12
Adult	SVM (RBF)	33	25	25	25	25	12
Adult	SVM (RBF)	44	25	25	25	25	12
Adult	SVM (RBF)	55	25	25	25	25	12
Adult	MLP	11	25	25	25	25	12
Adult	MLP	22	25	25	25	25	12
Adult	MLP	33	25	25	25	25	12
Adult	MLP	44	25	25	25	25	12
Adult	MLP	55	25	25	25	25	12

Dataset	Model	Seed	TP	TN	FP	FN	Stability
Breast Cancer	Logistic regression	11	25	25	1	1	8
Breast Cancer	Logistic regression	22	25	25	1	1	8
Breast Cancer	Logistic regression	33	25	25	1	1	8
Breast Cancer	Random forest	11	25	25	3	2	11
Breast Cancer	Random forest	22	25	25	3	2	11
Breast Cancer	Random forest	33	25	25	3	2	11
Breast Cancer	XGBoost	11	25	25	4	2	11
Breast Cancer	XGBoost	22	25	25	4	2	11
Breast Cancer	XGBoost	33	25	25	4	2	11
Breast Cancer	SVM (RBF)	11	25	25	1	1	8
Breast Cancer	SVM (RBF)	22	25	25	1	1	8
Breast Cancer	SVM (RBF)	33	25	25	1	1	8
Breast Cancer	MLP	11	25	25	1	3	10
Breast Cancer	MLP	22	25	25	1	3	10
Breast Cancer	MLP	33	25	25	1	3	10
German Credit	Logistic regression	11	25	25	25	25	12
German Credit	Logistic regression	22	25	25	25	25	12
German Credit	Logistic regression	33	25	25	25	25	12
German Credit	Random forest	11	25	25	25	18	12

Dataset	Model	Seed	TP	TN	FP	FN	Stability
German Credit	Random forest	22	25	25	25	18	12
German Credit	Random forest	33	25	25	25	18	12
German Credit	XGBoost	11	25	25	25	21	12
German Credit	XGBoost	22	25	25	25	21	12
German Credit	XGBoost	33	25	25	25	21	12
German Credit	SVM (RBF)	11	25	23	25	15	12
German Credit	SVM (RBF)	22	25	23	25	15	12
German Credit	SVM (RBF)	33	25	23	25	15	12
German Credit	MLP	11	25	25	25	25	12
German Credit	MLP	22	25	25	25	25	12
German Credit	MLP	33	25	25	25	25	12
Diabetes	Logistic regression	11	25	25	18	25	12
Diabetes	Logistic regression	22	25	25	18	25	12
Diabetes	Logistic regression	33	25	25	18	25	12
Diabetes	Random forest	11	25	25	15	23	12
Diabetes	Random forest	22	25	25	15	23	12
Diabetes	Random forest	33	25	25	15	23	12
Diabetes	XGBoost	11	25	25	16	20	12
Diabetes	XGBoost	22	25	25	16	20	12
Diabetes	XGBoost	33	25	25	16	20	12
Diabetes	SVM (RBF)	11	25	25	17	21	12
Diabetes	SVM (RBF)	22	25	25	17	21	12
Diabetes	SVM (RBF)	33	25	25	17	21	12
Diabetes	MLP	11	25	25	19	25	12
Diabetes	MLP	22	25	25	19	25	12

Dataset	Model	Seed	TP	TN	FP	FN	Stability
Diabetes	MLP	33	25	25	19	25	12
Spambase	Logistic regression	11	25	25	25	25	12
Spambase	Logistic regression	22	25	25	25	25	12
Spambase	Logistic regression	33	25	25	25	25	12
Spambase	Random forest	11	25	25	18	25	12
Spambase	Random forest	22	25	25	18	25	12
Spambase	Random forest	33	25	25	18	25	12
Spambase	XGBoost	11	25	25	24	25	12
Spambase	XGBoost	22	25	25	24	25	12
Spambase	XGBoost	33	25	25	24	25	12
Spambase	SVM (RBF)	11	25	25	25	25	12
Spambase	SVM (RBF)	22	25	25	25	25	12
Spambase	SVM (RBF)	33	25	25	25	25	12
Spambase	MLP	11	25	25	25	25	12
Spambase	MLP	22	25	25	25	25	12
Spambase	MLP	33	25	25	25	25	12
Phishing	Logistic regression	11	25	25	25	25	12
Phishing	Logistic regression	22	25	25	25	25	12
Phishing	Logistic regression	33	25	25	25	25	12
Phishing	Random forest	11	25	25	25	24	12
Phishing	Random forest	22	25	25	25	24	12
Phishing	Random forest	33	25	25	25	24	12
Phishing	XGBoost	11	25	25	25	25	12
Phishing	XGBoost	22	25	25	25	25	12
Phishing	XGBoost	33	25	25	25	25	12
Phishing	SVM (RBF)	11	25	25	25	25	12
Phishing	SVM (RBF)	22	25	25	25	25	12
Phishing	SVM (RBF)	33	25	25	25	25	12
Phishing	MLP	11	25	25	25	22	12
Phishing	MLP	22	25	25	25	22	12
Phishing	MLP	33	25	25	25	22	12
Connect-4	Logistic regression	11	25	25	25	25	12
Connect-4	Logistic regression	22	25	25	25	25	12
Connect-4	Logistic regression	33	25	25	25	25	12

Dataset	Model	Seed	TP	TN	FP	FN	Stability
Connect-4	Random forest	11	25	25	25	25	12
Connect-4	Random forest	22	25	25	25	25	12
Connect-4	Random forest	33	25	25	25	25	12
Connect-4	XGBoost	11	25	25	25	25	12
Connect-4	XGBoost	22	25	25	25	25	12
Connect-4	XGBoost	33	25	25	25	25	12
Connect-4	SVM (RBF)	11	25	25	25	25	12
Connect-4	SVM (RBF)	22	25	25	25	25	12
Connect-4	SVM (RBF)	33	25	25	25	25	12
Connect-4	MLP	11	25	25	25	25	12
Connect-4	MLP	22	25	25	25	25	12
Connect-4	MLP	33	25	25	25	25	12

Overlap of the instance sets between seeds (Jaccard index over all pairs of seeds of the same dataset and model):

Dataset	Model	Mean	Maximum
Adult	Logistic regression	0.01	0.04
Adult	Random forest	0.01	0.02
Adult	XGBoost	0.02	0.03
Adult	SVM (RBF)	0.01	0.03
Adult	MLP	0.01	0.02
Breast Cancer	Logistic regression	0.32	0.37
Breast Cancer	Random forest	0.39	0.41
Breast Cancer	XGBoost	0.39	0.42
Breast Cancer	SVM (RBF)	0.32	0.37
Breast Cancer	MLP	0.33	0.35
German Credit	Logistic regression	0.54	0.55
German Credit	Random forest	0.53	0.55
German Credit	XGBoost	0.53	0.55
German Credit	SVM (RBF)	0.53	0.54
German Credit	MLP	0.55	0.57
Diabetes	Logistic regression	0.66	0.69
Diabetes	Random forest	0.57	0.57
Diabetes	XGBoost	0.55	0.58

Dataset	Model	Mean	Maximum
Diabetes	SVM (RBF)	0.58	0.61
Diabetes	MLP	0.65	0.66
Spambase	Logistic regression	0.29	0.31
Spambase	Random forest	0.26	0.27
Spambase	XGBoost	0.34	0.35
Spambase	SVM (RBF)	0.29	0.31
Spambase	MLP	0.31	0.32
Phishing	Logistic regression	0.12	0.14
Phishing	Random forest	0.26	0.27
Phishing	XGBoost	0.23	0.24
Phishing	SVM (RBF)	0.19	0.20
Phishing	MLP	0.26	0.27
Connect-4	Logistic regression	0.00	0.01
Connect-4	Random forest	0.01	0.02
Connect-4	XGBoost	0.01	0.01
Connect-4	SVM (RBF)	0.01	0.02
Connect-4	MLP	0.02	0.03

Over all 140 pairs of seeds: mean 0.22, maximum 0.69 (the values of §3.2 of the paper).

Degenerate explanations per dataset and explainer: share of all-zero attribution vectors, share of instances with undefined fidelity, and share of all-zero vectors among the 15 perturbed explanations of each stability instance:

Dataset	Explainer	Rows	All-zero	Fidelity undefined	Zero among perturbed
Adult	SHAP	2,500	0.0%	0.0%	0.0%
Adult	LIME (10)	2,500	0.4%	0.0%	5.8%
Adult	LIME (full)	2,500	0.4%	0.0%	6.3%
Adult	Anchors	2,498	0.0%	0.1%	0.0%
Adult	DiCE	2,500	0.0%	0.0%	0.0%
Breast Cancer	SHAP	807	0.0%	3.3%	0.0%
Breast Cancer	LIME (10)	807	0.0%	3.3%	0.0%
Breast Cancer	LIME (full)	807	0.0%	3.3%	0.0%

Dataset	Explainer	Rows	All-zero	Fidelity undefined	Zero among perturbed
Breast Cancer	Anchors	807	0.0%	3.3%	0.0%
Breast Cancer	DiCE	807	5.9%	8.7%	3.1%
German Credit	SHAP	1,431	0.0%	0.0%	0.0%
German Credit	LIME (10)	1,431	0.0%	0.0%	0.0%
German Credit	LIME (full)	1,431	0.0%	0.0%	0.0%
German Credit	Anchors	1,431	0.0%	0.0%	0.0%
German Credit	DiCE	1,431	0.0%	0.0%	0.0%
Diabetes	SHAP	1,347	0.0%	0.0%	0.0%
Diabetes	LIME (10)	1,347	0.0%	0.0%	0.0%
Diabetes	LIME (full)	1,347	0.0%	0.0%	0.0%
Diabetes	Anchors	1,347	0.0%	1.4%	0.0%
Diabetes	DiCE	1,347	0.0%	0.0%	0.0%
Spambase	SHAP	1,476	0.0%	1.2%	0.0%
Spambase	LIME (10)	1,476	0.3%	1.2%	0.1%
Spambase	LIME (full)	1,476	0.3%	1.2%	0.1%
Spambase	Anchors	1,476	0.0%	1.2%	0.0%
Spambase	DiCE	1,476	0.6%	1.8%	0.5%
Phishing	SHAP	1,488	0.0%	1.4%	0.0%
Phishing	LIME (10)	1,488	0.0%	1.4%	0.0%
Phishing	LIME (full)	1,488	0.0%	1.4%	0.0%
Phishing	Anchors	1,488	0.0%	1.4%	0.0%
Phishing	DiCE	1,488	0.0%	1.4%	0.0%
Connect-4	SHAP	1,500	0.0%	0.3%	0.0%
Connect-4	LIME (10)	1,500	0.0%	0.3%	0.8%
Connect-4	LIME (full)	1,500	0.0%	0.3%	0.8%
Connect-4	Anchors	1,497	0.0%	0.3%	0.0%
Connect-4	DiCE	1,500	0.0%	0.3%	0.0%

S8. Primary benchmark (Adult): per model and seed

Mean of each endpoint per cell (model, seed) and explainer. LIME (10) uses the 10-feature budget, LIME (full) the full vector; the confirmatory contrasts use LIME (full).

Model	Seed	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
Logistic regression	11	SHAP	0.88	0.96	0.33	0.09
Logistic regression	11	LIME (10)	0.45	0.61	0.26	0.09
Logistic regression	11	LIME (full)	0.61	0.57	0.25	0.98
Logistic regression	11	Anchors	0.63	0.54	0.25	0.03
Logistic regression	11	DiCE	0.41	0.16	0.06	0.02
Logistic regression	22	SHAP	0.88	0.95	0.35	0.09
Logistic regression	22	LIME (10)	0.35	0.62	0.28	0.09
Logistic regression	22	LIME (full)	0.62	0.59	0.28	0.98
Logistic regression	22	Anchors	0.61	0.49	0.22	0.04
Logistic regression	22	DiCE	0.35	0.20	0.07	0.02
Logistic regression	33	SHAP	0.84	0.96	0.35	0.09
Logistic regression	33	LIME (10)	0.42	0.62	0.28	0.09
Logistic regression	33	LIME (full)	0.58	0.59	0.27	0.99
Logistic regression	33	Anchors	0.66	0.55	0.25	0.03
Logistic regression	33	DiCE	0.38	0.17	0.06	0.02
Logistic regression	44	SHAP	0.84	0.96	0.35	0.09
Logistic regression	44	LIME (10)	0.48	0.61	0.29	0.09

Model	Seed	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
Logistic regression	44	LIME (full)	0.62	0.58	0.27	0.97
Logistic regression	44	Anchors	0.64	0.51	0.22	0.04
Logistic regression	44	DiCE	0.30	0.18	0.06	0.02
Logistic regression	55	SHAP	0.86	0.96	0.36	0.09
Logistic regression	55	LIME (10)	0.44	0.65	0.30	0.09
Logistic regression	55	LIME (full)	0.57	0.62	0.29	0.98
Logistic regression	55	Anchors	0.60	0.50	0.24	0.03
Logistic regression	55	DiCE	0.32	0.20	0.07	0.02
Random forest	11	SHAP	0.99	0.81	0.35	0.44
Random forest	11	LIME (10)	0.38	0.36	0.26	0.09
Random forest	11	LIME (full)	0.54	0.33	0.24	0.98
Random forest	11	Anchors	0.66	0.43	0.20	0.05
Random forest	11	DiCE	0.61	0.13	0.06	0.02
Random forest	22	SHAP	0.96	0.79	0.34	0.44
Random forest	22	LIME (10)	0.46	0.35	0.28	0.09
Random forest	22	LIME (full)	0.64	0.31	0.24	0.96
Random forest	22	Anchors	0.60	0.40	0.20	0.05
Random forest	22	DiCE	0.43	0.17	0.08	0.02
Random forest	33	SHAP	0.98	0.81	0.34	0.44
Random forest	33	LIME (10)	0.34	0.37	0.26	0.09
Random forest	33	LIME (full)	0.58	0.33	0.23	0.95
Random forest	33	Anchors	0.66	0.40	0.21	0.06
Random forest	33	DiCE	0.43	0.16	0.06	0.02
Random forest	44	SHAP	0.99	0.84	0.38	0.44
Random forest	44	LIME (10)	0.34	0.38	0.29	0.09
Random forest	44	LIME (full)	0.59	0.34	0.25	0.97
Random forest	44	Anchors	0.75	0.42	0.23	0.04
Random forest	44	DiCE	0.53	0.14	0.06	0.02

Model	Seed	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
Random forest	55	SHAP	0.99	0.83	0.36	0.44
Random forest	55	LIME (10)	0.35	0.36	0.27	0.09
Random forest	55	LIME (full)	0.50	0.33	0.25	0.99
Random forest	55	Anchors	0.62	0.39	0.18	0.04
Random forest	55	DiCE	0.48	0.15	0.07	0.02
XGBoost	11	SHAP	0.63	0.76	0.45	0.36
XGBoost	11	LIME (10)	0.28	0.62	0.44	0.09
XGBoost	11	LIME (full)	0.54	0.58	0.43	0.99
XGBoost	11	Anchors	0.58	0.33	0.22	0.05
XGBoost	11	DiCE	0.50	0.35	0.17	0.02
XGBoost	22	SHAP	0.63	0.77	0.41	0.36
XGBoost	22	LIME (10)	0.22	0.60	0.41	0.09
XGBoost	22	LIME (full)	0.52	0.57	0.39	0.98
XGBoost	22	Anchors	0.59	0.32	0.20	0.07
XGBoost	22	DiCE	0.71	0.32	0.16	0.02
XGBoost	33	SHAP	0.61	0.76	0.44	0.36
XGBoost	33	LIME (10)	0.19	0.64	0.43	0.09
XGBoost	33	LIME (full)	0.63	0.60	0.41	0.99
XGBoost	33	Anchors	0.50	0.32	0.21	0.05
XGBoost	33	DiCE	0.59	0.34	0.16	0.02
XGBoost	44	SHAP	0.65	0.77	0.45	0.36
XGBoost	44	LIME (10)	0.27	0.63	0.44	0.09
XGBoost	44	LIME (full)	0.57	0.60	0.43	0.99
XGBoost	44	Anchors	0.59	0.34	0.21	0.07
XGBoost	44	DiCE	0.58	0.29	0.18	0.02
XGBoost	55	SHAP	0.56	0.78	0.42	0.36
XGBoost	55	LIME (10)	0.25	0.62	0.42	0.09
XGBoost	55	LIME (full)	0.51	0.59	0.42	0.99
XGBoost	55	Anchors	0.57	0.31	0.18	0.07
XGBoost	55	DiCE	0.53	0.27	0.15	0.02
SVM (RBF)	11	SHAP	0.91	0.89	0.35	0.09
SVM (RBF)	11	LIME (10)	0.33	0.50	0.23	0.09
SVM (RBF)	11	LIME (full)	0.54	0.47	0.19	0.99
SVM (RBF)	11	Anchors	0.67	0.53	0.28	0.05

Model	Seed	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
SVM (RBF)	11	DiCE	0.46	0.22	0.06	0.01
SVM (RBF)	22	SHAP	0.93	0.90	0.35	0.09
SVM (RBF)	22	LIME (10)	0.40	0.51	0.25	0.09
SVM (RBF)	22	LIME (full)	0.61	0.48	0.22	0.99
SVM (RBF)	22	Anchors	0.74	0.52	0.30	0.04
SVM (RBF)	22	DiCE	0.53	0.27	0.08	0.02
SVM (RBF)	33	SHAP	0.90	0.89	0.35	0.09
SVM (RBF)	33	LIME (10)	0.44	0.55	0.24	0.09
SVM (RBF)	33	LIME (full)	0.60	0.52	0.20	0.99
SVM (RBF)	33	Anchors	0.65	0.49	0.26	0.04
SVM (RBF)	33	DiCE	0.31	0.25	0.06	0.02
SVM (RBF)	44	SHAP	0.90	0.90	0.36	0.09
SVM (RBF)	44	LIME (10)	0.40	0.54	0.26	0.09
SVM (RBF)	44	LIME (full)	0.61	0.51	0.22	0.99
SVM (RBF)	44	Anchors	0.68	0.49	0.29	0.04
SVM (RBF)	44	DiCE	0.39	0.22	0.06	0.01
SVM (RBF)	55	SHAP	0.88	0.89	0.35	0.09
SVM (RBF)	55	LIME (10)	0.42	0.51	0.23	0.09
SVM (RBF)	55	LIME (full)	0.62	0.49	0.20	0.98
SVM (RBF)	55	Anchors	0.70	0.57	0.28	0.04
SVM (RBF)	55	DiCE	0.37	0.23	0.06	0.01
MLP	11	SHAP	0.26	0.72	0.49	0.09
MLP	11	LIME (10)	0.18	0.44	0.30	0.09
MLP	11	LIME (full)	0.45	0.23	0.20	0.99
MLP	11	Anchors	0.63	0.31	0.28	0.12
MLP	11	DiCE	0.11	0.02	0.04	0.02
MLP	22	SHAP	0.38	0.74	0.53	0.09
MLP	22	LIME (10)	0.21	0.44	0.32	0.09
MLP	22	LIME (full)	0.45	0.22	0.19	1.00
MLP	22	Anchors	0.65	0.34	0.27	0.12
MLP	22	DiCE	0.16	0.02	0.04	0.02
MLP	33	SHAP	0.38	0.74	0.52	0.09
MLP	33	LIME (10)	0.20	0.43	0.30	0.09
MLP	33	LIME (full)	0.48	0.23	0.19	0.97

Model	Seed	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
MLP	33	Anchors	0.62	0.27	0.22	0.15
MLP	33	DiCE	0.20	0.03	0.03	0.02
MLP	44	SHAP	0.28	0.72	0.49	0.09
MLP	44	LIME (10)	0.17	0.47	0.30	0.09
MLP	44	LIME (full)	0.46	0.25	0.19	0.98
MLP	44	Anchors	0.62	0.27	0.21	0.13
MLP	44	DiCE	0.13	0.05	0.07	0.02
MLP	55	SHAP	0.32	0.69	0.42	0.09
MLP	55	LIME (10)	0.23	0.46	0.30	0.09
MLP	55	LIME (full)	0.47	0.24	0.16	0.98
MLP	55	Anchors	0.62	0.26	0.24	0.13
MLP	55	DiCE	0.12	0.01	0.04	0.02

Paired effects SHAP − X on Adult, every endpoint and contrast (confirmatory rows marked C; Holm correction over the 12 confirmatory tests). Mixed model with random intercepts for model and for seed within model; variance components of the model (between models, seeds within models, residual); cluster bootstrap over models; number of the 25 cells with a positive difference.

Endpoint	X		Estimate [95% CI]	p	p Holm	Bootstrap 95% CI	Var. model	Var. seed	Var. residual	Cells > 0
Stability	LIME (full)	C	0.181 [−0.010, 0.372]	0.063	0.253	[−0.008, 0.339]	0.0465	0.0000	0.0506	19
Stability	Anchors	C	0.102 [−0.078, 0.283]	0.266	0.312	[−0.108, 0.264]	0.0417	0.0012	0.0317	19
Stability	DiCE	C	0.339 [0.136, 0.543]	0.001	0.008	[0.153, 0.498]	0.0521	0.0002	0.0965	24
Stability	LIME (10)		0.408 [0.247, 0.570]	<0.001	—	[0.251, 0.537]	0.0331	0.0017	0.0375	25
Fidelity	LIME (full)	C	0.384 [0.225, 0.544]	<0.001	<0.001	[0.278, 0.470]	0.0331	0.0000	0.0425	25

Endpoint	X		Estimate [95% CI]	p	p Holm	Bootstrap 95% CI	Var. model	Var. seed	Var. residual	Cells > 0
Fidelity	Anchors	C	0.419 [0.253, 0.585]	<0.001	<0.001	[0.395, 0.439]	0.0356	0.0003	0.0523	25
Fidelity	DiCE	C	0.649 [0.440, 0.858]	<0.001	<0.001	[0.544, 0.729]	0.0568	0.0001	0.0776	25
Fidelity	LIME (10)		0.314 [0.153, 0.476]	<0.001	—	[0.221, 0.399]	0.0338	0.0000	0.0444	25
Faithfulness gap	LIME (full)	C	0.131 [0.051, 0.212]	0.001	0.008	[0.057, 0.221]	0.0083	0.0000	0.0341	25
Faithfulness gap	Anchors	C	0.161 [−0.013, 0.334]	0.069	0.253	[0.101, 0.220]	0.0391	0.0001	0.0547	25
Faithfulness gap	DiCE	C	0.315 [0.127, 0.502]	<0.001	0.008	[0.276, 0.381]	0.0455	0.0001	0.0643	25
Faithfulness gap	LIME (10)		0.091 [−0.034, 0.217]	0.154	—	[0.042, 0.143]	0.0204	0.0000	0.0282	24
Sparsity	LIME (full)		−0.766 [−0.876, −0.657]	<0.001	—	[−0.891, −0.624]	0.0157	0.0000	0.0080	0
Sparsity	Anchors		0.151 [0.032, 0.271]	0.013	—	[0.016, 0.304]	0.0185	0.0000	0.0124	20
Sparsity	DiCE		0.198 [−0.151, 0.548]	0.266	—	[0.077, 0.335]	0.1590	0.0000	0.0004	25
Sparsity	LIME (10)		0.123 [−0.031, 0.277]	0.116	—	[0.001, 0.261]	0.0308	0.0000	0.0003	21
Wall time (ms)	LIME (full	C	3,756 [561, 6,950]	0.021	0.106	[−0, 11,244]	13,282,976	235.1689	9,937	19

Endpoint	X		Estimate [95% CI]	p	p Holm	Bootstrap 95% CI	Var. model	Var. seed	Var. residual	Cells > 0
	)									
Wall time (ms)	Anchors	C	-5,357 [-7,208, -3,507]	<0.001	<0.001	[-7,854, -2,861]	725.9988	3,507,119	375,329,671	0
Wall time (ms)	DiCE	C	2,517 [−961, 5,995]	0.156	0.312	[-1,094, 9,702]	15,745,746	3,142	10,603	5
Wall time (ms)	LIME (10)		3,757 [544, 6,971]	0.022	—	[1, 11,246]	13,440,870	314.8729	9,953	20
CPU time (ms)	LIME (full)		3,726 [-3,558, 11,009]	0.316	—	[−2, 11,159]	69,042,926	368.1309	5,255	18
CPU time (ms)	Anchors		-5,314 [-8,156, -2,471]	<0.001	—	[-7,789, -2,838]	6,188,102	3,358,922	365,431,338	0
CPU time (ms)	DiCE		2,496 [−415, 5,407]	0.093	—	[-1,088, 9,631]	11,027,902	5.1734	6,459	5
CPU time (ms)	LIME (10)		4,002 [-675,519,485, 675,527,489]	1.0000	—	[−1, 11,162]	61,751,436	0.0000	5,803	20
Rows evaluated	LIME (full)		11,561 [10,413, 12,708]	<0.001	—	[4,000, 24,001]	1,626,351	19.4509	0.0000	15
Rows evaluated	Anchors		-15,004 [-55,272, 25,263]	0.465	—	[-29,767, −242]	2,071,701,154	41,208,886	3,053,769,993	5
Rows evaluated	DiCE		12,958 [7,048, 18,868]	<0.001	—	[2,930, 22,984]	45,458,742	14,769	80,076	15
Rows evaluated	LIME (10)		11,561 [10,413, 12,708]	<0.001	—	[4,000, 24,001]	1,626,351	19.4509	0.0000	15

Distribution of the instance-level paired differences SHAP − X (quantiles over all Adult instances, and per model):

Endpoint	X	Model	n	Mean	Share > 0	Min	P10	Q1	Median	Q3	P90	Max
Stability	LIME (full)	All	300	0.18	73%	−0.60	−0.29	−0.02	0.24	0.41	0.52	0.80
Stability	LIME (full)	Logistic regression	60	0.26	93%	−0.27	0.04	0.14	0.28	0.40	0.45	0.80
Stability	LIME (full)	MLP	60	−0.14	30%	−0.60	−0.42	−0.36	−0.17	0.04	0.23	0.54
Stability	LIME (full)	Random forest	60	0.41	100%	0.11	0.18	0.29	0.42	0.53	0.60	0.73
Stability	LIME (full)	SVM (RBF)	60	0.31	92%	−0.32	0.02	0.20	0.34	0.42	0.52	0.71
Stability	LIME (full)	XGBoost	60	0.06	52%	−0.47	−0.29	−0.16	0.02	0.32	0.43	0.66
Stability	Anchors	All	298	0.10	71%	−0.81	−0.35	−0.06	0.20	0.32	0.38	0.66
Stability	Anchors	Logistic regression	60	0.23	93%	−0.23	0.05	0.15	0.25	0.33	0.38	0.47
Stability	Anchors	MLP	60	−0.30	12%	−0.81	−0.55	−0.45	−0.35	−0.14	0.01	0.19
Stability	Anchors	Random forest	60	0.32	100%	0.02	0.15	0.26	0.34	0.38	0.44	0.66
Stability	Anchors	SVM (RBF)	58	0.21	88%	−0.24	−0.02	0.11	0.26	0.33	0.39	0.45
Stability	Anchors	XGBoost	60	0.05	65%	−0.39	−0.22	−0.13	0.08	0.22	0.27	0.47
Stability	DiCE	All	300	0.34	81%	−0.64	−0.06	0.03	0.29	0.71	0.84	0.96
Stability	DiCE	Logistic regression	60	0.51	83%	−0.09	−0.05	0.18	0.71	0.80	0.84	0.88
Stability	DiCE	MLP	60	0.18	83%	−0.36	−0.03	0.05	0.12	0.30	0.48	0.82
Stability	DiCE	Random forest	60	0.49	98%	−0.03	0.01	0.09	0.63	0.80	0.88	0.96
Stability	DiCE	SVM (RBF)	60	0.49	83%	−0.13	−0.04	0.17	0.57	0.82	0.87	0.93

Endpoint	X	Model	n	Mean	Share > 0	Min	P10	Q1	Median	Q3	P90	Max
Stability	DiCE	XGBoost	60	0.03	55%	−0.64	−0.32	−0.14	0.02	0.21	0.38	0.55
Fidelity	LIME (full)	All	2,500	0.38	95%	−0.63	0.05	0.23	0.41	0.55	0.65	1.08
Fidelity	LIME (full)	Logistic regression	500	0.37	100%	0.01	0.21	0.32	0.39	0.44	0.50	0.77
Fidelity	LIME (full)	MLP	500	0.49	98%	−0.45	0.22	0.34	0.47	0.64	0.77	1.08
Fidelity	LIME (full)	Random forest	500	0.49	99%	−0.23	0.21	0.45	0.52	0.60	0.64	0.90
Fidelity	LIME (full)	SVM (RBF)	500	0.40	96%	−0.16	0.05	0.17	0.47	0.60	0.66	0.91
Fidelity	LIME (full)	XGBoost	500	0.18	82%	−0.63	−0.09	0.03	0.18	0.29	0.53	0.83
Fidelity	Anchors	All	2,498	0.42	97%	−0.59	0.13	0.27	0.41	0.58	0.72	1.25
Fidelity	Anchors	Logistic regression	500	0.44	100%	0.03	0.19	0.31	0.42	0.58	0.70	0.96
Fidelity	Anchors	MLP	500	0.43	95%	−0.59	0.13	0.28	0.44	0.59	0.71	1.25
Fidelity	Anchors	Random forest	500	0.41	94%	−0.31	0.08	0.26	0.39	0.58	0.78	0.96
Fidelity	Anchors	SVM (RBF)	498	0.37	98%	−0.53	0.12	0.22	0.36	0.51	0.66	0.95
Fidelity	Anchors	XGBoost	500	0.44	97%	−0.29	0.14	0.29	0.46	0.61	0.74	0.98
Fidelity	DiCE	All	2,500	0.65	97%	−0.63	0.21	0.45	0.73	0.90	0.96	1.02
Fidelity	DiCE	Logistic regression	500	0.77	100%	−0.00	0.45	0.67	0.81	0.96	0.99	1.02
Fidelity	DiCE	MLP	500	0.69	98%	−0.14	0.42	0.59	0.74	0.85	0.92	1.01
Fidelity	DiCE	Random forest	500	0.66	100%	−0.07	0.25	0.41	0.79	0.91	0.94	1.00
Fidelity	DiCE	SVM (RBF)	500	0.66	99%	−0.04	0.20	0.39	0.73	0.93	0.98	1.01

Endpoint	X	Model	n	Mean	Share > 0	Min	P10	Q1	Median	Q3	P90	Max
Fidelity	DiCE	XGBoost	500	0.45	88%	−0.63	−0.02	0.22	0.49	0.74	0.87	0.99
Faithfulness gap	LIME (full)	All	2,500	0.13	74%	−0.65	−0.03	0.00	0.06	0.21	0.39	1.00
Faithfulness gap	LIME (full)	Logistic regression	500	0.08	73%	−0.22	−0.07	−0.00	0.06	0.15	0.24	0.57
Faithfulness gap	LIME (full)	MLP	500	0.30	86%	−0.65	−0.04	0.04	0.26	0.56	0.78	1.00
Faithfulness gap	LIME (full)	Random forest	500	0.12	87%	−0.09	−0.00	0.02	0.11	0.20	0.25	0.40
Faithfulness gap	LIME (full)	SVM (RBF)	500	0.14	81%	−0.26	−0.04	0.01	0.08	0.26	0.39	0.79
Faithfulness gap	LIME (full)	XGBoost	500	0.02	42%	−0.08	−0.03	−0.01	0.00	0.01	0.03	0.88
Faithfulness gap	Anchors	All	2,498	0.16	80%	−0.81	−0.06	0.01	0.11	0.28	0.48	0.99
Faithfulness gap	Anchors	Logistic regression	500	0.11	75%	−0.35	−0.08	−0.00	0.09	0.21	0.35	0.77
Faithfulness gap	Anchors	MLP	500	0.24	80%	−0.81	−0.17	0.02	0.21	0.48	0.72	0.98
Faithfulness gap	Anchors	Random forest	500	0.15	89%	−0.19	0.00	0.03	0.15	0.23	0.35	0.73
Faithfulness gap	Anchors	SVM (RBF)	498	0.07	71%	−0.58	−0.11	−0.00	0.05	0.12	0.27	0.86
Faithfulness gap	Anchors	XGBoost	500	0.23	86%	−0.32	−0.00	0.02	0.15	0.38	0.56	0.99
Faithfulness gap	DiCE	All	2,500	0.31	93%	−0.80	0.01	0.07	0.30	0.50	0.68	0.99
Faithfulness gap	DiCE	Logistic regression	500	0.28	93%	−0.25	0.02	0.09	0.27	0.46	0.58	0.80
Faithfulness gap	DiCE	MLP	500	0.44	95%	−0.80	0.02	0.16	0.45	0.73	0.89	0.99
Faithfulness gap	DiCE	Random forest	500	0.29	98%	−0.08	0.02	0.12	0.31	0.43	0.53	0.74
Faithfulness gap	DiCE	SVM (RBF)	500	0.29	90%	−0.49	−0.00	0.04	0.25	0.53	0.68	0.86

Endpoint	X	Model	n	Mean	Share > 0	Min	P10	Q1	Median	Q3	P90	Max
Faithfulness gap	DiCE	XGBoost	500	0.27	89%	−0.16	−0.00	0.02	0.24	0.46	0.61	0.99

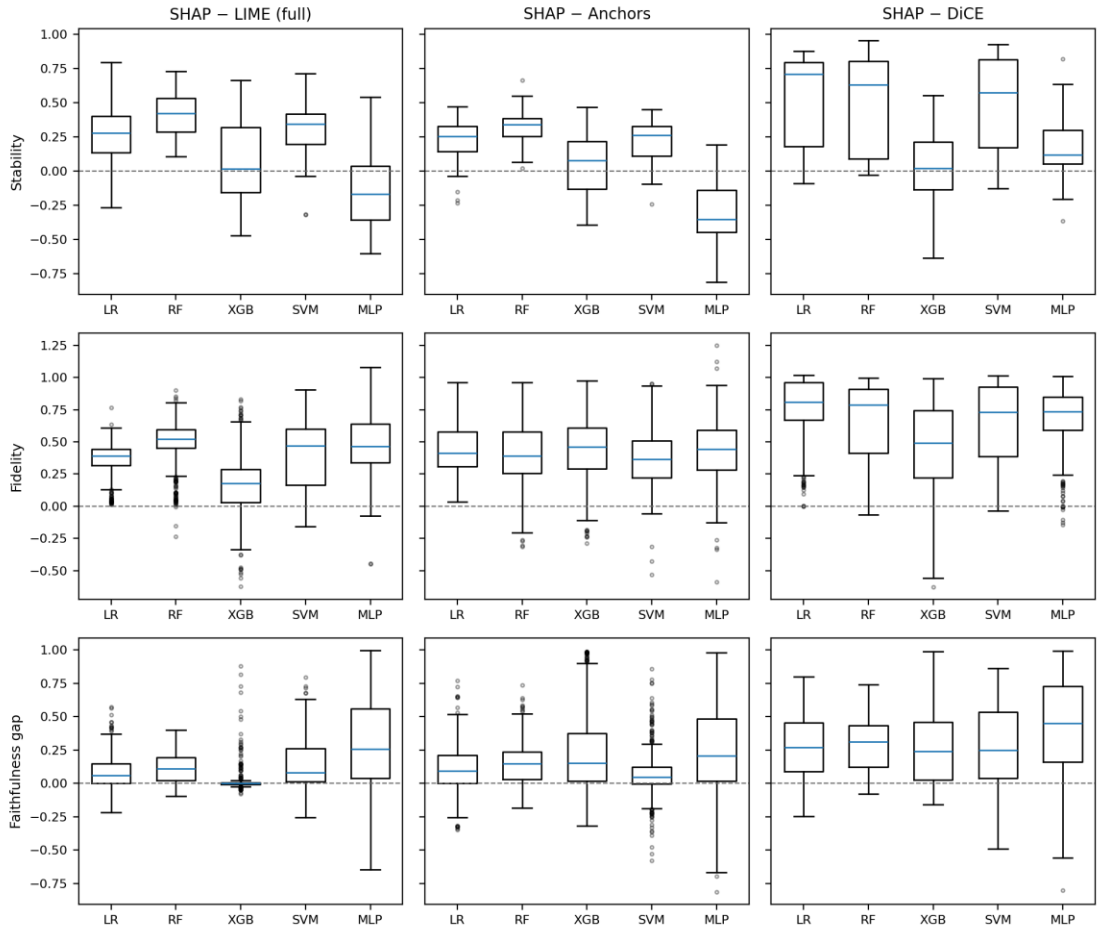


Figure S1. Instance-level paired differences SHAP − X on Adult, by model (LR logistic regression, RF random forest, XGB XGBoost, SVM, MLP). Dashed line: no difference.

S9. Cost

Timing pass: 5 instances per stratum (20 per cell), 3 repetitions (median used), 1 warm-up explanation per explainer, explainers interleaved per instance, one process and one thread pinned to logical CPU 2, measurements paused while any other process used more than 10% of a core. Wall time from the high-resolution performance counter; CPU time from the process clock (15.6 ms steps on Windows). FLOPs per explanation for logistic regression (2p + 2 per row) and the MLP (one hidden layer of 100 units), with p the number of encoded features.

Dataset	Model	Explainer	n	Wall (ms)	CPU (ms)	Rows	Model calls	MFLOPs
Adult	Logistic regression	SHAP	100	29.6	31.2	25,001	2	5.45
Adult	Logistic regression	LIME (10)	100	7.8	15.6	1,000	1	0.22
Adult	Logistic regression	LIME (full)	100	9.4	15.6	1,000	1	0.22
Adult	Logistic regression	Anchors	100	748.9	742.2	5,501	56	1.20
Adult	Logistic regression	DiCE	100	1,064	1,062	2,004	6	0.44
Adult	Random forest	SHAP	100	23.0	15.6	0	0	—
Adult	Random forest	LIME (10)	100	20.5	15.6	1,000	1	—
Adult	Random forest	LIME (full)	100	22.3	15.6	1,000	1	—
Adult	Random forest	Anchors	100	1,323	1,320	6,551	66	—
Adult	Random forest	DiCE	100	1,107	1,109	2,004	6	—
Adult	XGBoost	SHAP	100	1.4	0.0	0	0	—
Adult	XGBoost	LIME (10)	100	11.5	15.6	1,000	1	—
Adult	XGBoost	LIME (full)	100	13.3	15.6	1,000	1	—
Adult	XGBoost	Anchors	100	918.8	921.9	6,901	70	—
Adult	XGBoost	DiCE	100	1,107	1,094	2,004	6	—
Adult	SVM (RBF)	SHAP	100	19,492	19,359	25,001	2	—
Adult	SVM (RBF)	LIME (10)	100	788.5	781.2	1,000	1	—
Adult	SVM (RBF)	LIME (full)	100	786.2	781.2	1,000	1	—
Adult	SVM (RBF)	Anchors	100	5,640	5,617	6,351	64	—

Dataset	Model	Explainer	n	Wall (ms)	CPU (ms)	Rows	Model calls	MFLOPs
Adult	SVM (RBF)	DiCE	100	2,627	2,609	2,004	6	—
Adult	MLP	SHAP	100	52.2	46.9	25,001	2	555.02
Adult	MLP	LIME (10)	100	9.2	15.6	1,000	1	22.20
Adult	MLP	LIME (full)	100	11.0	15.6	1,000	1	22.20
Adult	MLP	Anchors	100	2,776	2,750	23,201	233	515.06
Adult	MLP	DiCE	100	1,094	1,094	2,004	6	44.49
Breast Cancer	Logistic regression	SHAP	36	18.0	15.6	25,001	2	1.55
Breast Cancer	Logistic regression	LIME (10)	36	3.8	0.0	1,000	1	0.06
Breast Cancer	Logistic regression	LIME (full)	36	3.8	0.0	1,000	1	0.06
Breast Cancer	Logistic regression	Anchors	36	1,534	1,523	64,001	641	3.97
Breast Cancer	Logistic regression	DiCE	36	228.4	234.4	2,004	6	0.12
Breast Cancer	Random forest	SHAP	45	2.6	0.0	0	0	—
Breast Cancer	Random forest	LIME (10)	45	11.0	15.6	1,000	1	—
Breast Cancer	Random forest	LIME (full)	45	11.1	15.6	1,000	1	—
Breast Cancer	Random forest	Anchors	45	5,367	5,328	86,501	866	—
Breast Cancer	Random forest	DiCE	45	294.9	296.9	2,004	6	—
Breast Cancer	XGBoost	SHAP	48	0.7	0.0	0	0	—
Breast Cancer	XGBoost	LIME (10)	48	5.5	0.0	1,000	1	—
Breast	XGBoost	LIME	48	5.6	0.0	1,00	1	—

Dataset	Model	Explainer	n	Wall (ms)	CPU (ms)	Rows	Model calls	MFLOPs
Cancer		(full)				0		
Breast Cancer	XGBoost	Anchors	48	2,071	2,047	70,401	705	—
Breast Cancer	XGBoost	DiCE	48	250.2	250.0	2,004	6	—
Breast Cancer	SVM (RBF)	SHAP	36	230.2	234.4	25,001	2	—
Breast Cancer	SVM (RBF)	LIME (10)	36	12.6	15.6	1,000	1	—
Breast Cancer	SVM (RBF)	LIME (full)	36	12.6	15.6	1,000	1	—
Breast Cancer	SVM (RBF)	Anchors	36	2,263	2,250	75,001	751	—
Breast Cancer	SVM (RBF)	DiCE	36	250.4	250.0	2,004	6	—
Breast Cancer	MLP	SHAP	42	32.8	31.2	25,001	2	165.01
Breast Cancer	MLP	LIME (10)	42	4.3	0.0	1,000	1	6.60
Breast Cancer	MLP	LIME (full)	42	4.2	0.0	1,000	1	6.60
Breast Cancer	MLP	Anchors	42	1,664	1,641	63,401	635	418.45
Breast Cancer	MLP	DiCE	42	233.2	234.4	2,004	6	13.23
German Credit	Logistic regression	SHAP	60	25.0	31.2	25,001	2	3.10
German Credit	Logistic regression	LIME (10)	60	5.2	0.0	1,000	1	0.12
German Credit	Logistic regression	LIME (full)	60	5.8	0.0	1,000	1	0.12
German Credit	Logistic regression	Anchors	60	848.2	851.6	19,302	194	2.39
German Credit	Logistic regression	DiCE	60	397.2	406.2	2,004	6	0.25
German Credit	Random forest	SHAP	60	13.3	15.6	0	0	—

Dataset	Model	Explainer	n	Wall (ms)	CPU (ms)	Rows	Model calls	MFLOPs
German Credit	Random forest	LIME (10)	60	16.0	15.6	1,000	1	—
German Credit	Random forest	LIME (full)	60	16.8	15.6	1,000	1	—
German Credit	Random forest	Anchors	60	6,099	6,047	77,551	776	—
German Credit	Random forest	DiCE	60	424.9	421.9	2,004	6	—
German Credit	XGBoost	SHAP	60	1.7	0.0	0	0	—
German Credit	XGBoost	LIME (10)	60	8.3	15.6	1,000	1	—
German Credit	XGBoost	LIME (full)	60	8.9	15.6	1,000	1	—
German Credit	XGBoost	Anchors	60	1,663	1,625	32,301	324	—
German Credit	XGBoost	DiCE	60	422.0	421.9	2,004	6	—
German Credit	SVM (RBF)	SHAP	60	1,165	1,156	25,001	2	—
German Credit	SVM (RBF)	LIME (10)	60	52.4	46.9	1,000	1	—
German Credit	SVM (RBF)	LIME (full)	60	53.0	46.9	1,000	1	—
German Credit	SVM (RBF)	Anchors	60	1,873	1,867	18,701	188	—
German Credit	SVM (RBF)	DiCE	60	488.1	484.4	2,004	6	—
German Credit	MLP	SHAP	60	42.7	46.9	25,001	2	320.01
German Credit	MLP	LIME (10)	60	6.0	0.0	1,000	1	12.80
German Credit	MLP	LIME (full)	60	6.6	0.0	1,000	1	12.80
German Credit	MLP	Anchors	60	1,850	1,836	46,352	466	593.31
German	MLP	DiCE	60	397.5	390.6	2,00	6	25.65

Dataset	Model	Explainer	n	Wall (ms)	CPU (ms)	Rows	Model calls	MFLOPs
Credit						4		
Diabetes	Logistic regression	SHAP	60	7.3	0.0	12,701	2	0.23
Diabetes	Logistic regression	LIME (10)	60	2.7	0.0	1,000	1	0.02
Diabetes	Logistic regression	LIME (full)	60	2.6	0.0	1,000	1	0.02
Diabetes	Logistic regression	Anchors	60	55.8	46.9	1,401	15	0.03
Diabetes	Logistic regression	DiCE	60	99.1	93.8	2,004	6	0.04
Diabetes	Random forest	SHAP	60	8.7	15.6	0	0	—
Diabetes	Random forest	LIME (10)	60	12.3	15.6	1,000	1	—
Diabetes	Random forest	LIME (full)	60	12.2	15.6	1,000	1	—
Diabetes	Random forest	Anchors	60	277.1	273.4	4,001	41	—
Diabetes	Random forest	DiCE	60	128.4	125.0	2,004	6	—
Diabetes	XGBoost	SHAP	60	1.6	0.0	0	0	—
Diabetes	XGBoost	LIME (10)	60	4.9	0.0	1,000	1	—
Diabetes	XGBoost	LIME (full)	60	4.9	0.0	1,000	1	—
Diabetes	XGBoost	Anchors	60	147.1	140.6	4,751	48.5	—
Diabetes	XGBoost	DiCE	60	105.1	109.4	2,004	6	—
Diabetes	SVM (RBF)	SHAP	60	316.9	312.5	12,701	2	—
Diabetes	SVM (RBF)	LIME (10)	60	27.6	31.2	1,000	1	—
Diabetes	SVM (RBF)	LIME (full)	60	27.5	31.2	1,000	1	—
Diabetes	SVM (RBF)	Anchors	60	120.9	117.2	2,00	21	—

Dataset	Model	Explainer	n	Wall (ms)	CPU (ms)	Rows	Model calls	MFLOPs
Diabetes	SVM (RBF)	DiCE	60	147.9	156.2	1 2,004	6	—
Diabetes	MLP	SHAP	60	14.1	15.6	12,701	2	27.94
Diabetes	MLP	LIME (10)	60	2.9	0.0	1,000	1	2.20
Diabetes	MLP	LIME (full)	60	2.9	0.0	1,000	1	2.20
Diabetes	MLP	Anchors	60	59.4	62.5	1,451	16	3.19
Diabetes	MLP	DiCE	60	99.9	93.8	2,004	6	4.41
Spambase	Logistic regression	SHAP	60	23.2	15.6	25,001	2	2.90
Spambase	Logistic regression	LIME (10)	60	5.2	0.0	1,000	1	0.12
Spambase	Logistic regression	LIME (full)	60	5.6	0.0	1,000	1	0.12
Spambase	Logistic regression	Anchors	60	1,086	1,086	29,751	298	3.45
Spambase	Logistic regression	DiCE	60	398.5	406.2	2,004	6	0.23
Spambase	Random forest	SHAP	60	14.3	15.6	0	0	—
Spambase	Random forest	LIME (10)	60	14.5	15.6	1,000	1	—
Spambase	Random forest	LIME (full)	60	15.0	15.6	1,000	1	—
Spambase	Random forest	Anchors	60	4,238	4,219	56,701	568	—
Spambase	Random forest	DiCE	60	432.2	437.5	2,004	6	—
Spambase	XGBoost	SHAP	60	1.3	0.0	0	0	—
Spambase	XGBoost	LIME (10)	60	8.0	15.6	1,000	1	—

Dataset	Model	Explainer	n	Wall (ms)	CPU (ms)	Rows	Model calls	MFLOPs
Spambase	XGBoost	LIME (full)	60	8.5	15.6	1,000	1	—
Spambase	XGBoost	Anchors	60	1,683	1,664	43,251	433.5	—
Spambase	XGBoost	DiCE	60	414.2	421.9	2,004	6	—
Spambase	SVM (RBF)	SHAP	60	2,310	2,297	25,001	2	—
Spambase	SVM (RBF)	LIME (10)	60	97.5	93.8	1,000	1	—
Spambase	SVM (RBF)	LIME (full)	60	98.0	93.8	1,000	1	—
Spambase	SVM (RBF)	Anchors	60	4,432	4,422	34,151	342	—
Spambase	SVM (RBF)	DiCE	60	590.1	593.8	2,004	6	—
Spambase	MLP	SHAP	60	40.3	46.9	25,001	2	300.01
Spambase	MLP	LIME (10)	60	5.7	0.0	1,000	1	12.00
Spambase	MLP	LIME (full)	60	6.1	0.0	1,000	1	12.00
Spambase	MLP	Anchors	60	1,305	1,281	36,751	368	441.01
Spambase	MLP	DiCE	60	401.5	406.2	2,004	6	24.05
Phishing	Logistic regression	SHAP	60	26.7	31.2	25,001	2	3.45
Phishing	Logistic regression	LIME (10)	60	5.9	0.0	1,000	1	0.14
Phishing	Logistic regression	LIME (full)	60	6.6	0.0	1,000	1	0.14
Phishing	Logistic regression	Anchors	60	1,746	1,742	29,151	292	4.02
Phishing	Logistic regression	DiCE	60	504.2	500.0	2,006	8	0.28
Phishing	Random	SHAP	60	23.3	15.6	0	0	—

Dataset	Model	Explainer	n	Wall (ms)	CPU (ms)	Rows	Model calls	MFLOPs
	forest							
Phishing	Random forest	LIME (10)	60	17.6	15.6	1,000	1	—
Phishing	Random forest	LIME (full)	60	18.3	15.6	1,000	1	—
Phishing	Random forest	Anchors	60	5,027	5,008	54,301	544	—
Phishing	Random forest	DiCE	60	531.1	531.2	2,005	7	—
Phishing	XGBoost	SHAP	60	1.7	0.0	0	0	—
Phishing	XGBoost	LIME (10)	60	8.9	15.6	1,000	1	—
Phishing	XGBoost	LIME (full)	60	9.6	15.6	1,000	1	—
Phishing	XGBoost	Anchors	60	2,070	2,023	29,151	292.5	—
Phishing	XGBoost	DiCE	60	524.2	531.2	2,006	8	—
Phishing	SVM (RBF)	SHAP	60	3,548	3,531	25,001	2	—
Phishing	SVM (RBF)	LIME (10)	60	146.9	140.6	1,000	1	—
Phishing	SVM (RBF)	LIME (full)	60	149.7	156.2	1,000	1	—
Phishing	SVM (RBF)	Anchors	60	8,099	8,070	39,951	400	—
Phishing	SVM (RBF)	DiCE	60	798.1	781.2	2,006	8	—
Phishing	MLP	SHAP	60	44.7	46.9	25,001	2	355.01
Phishing	MLP	LIME (10)	60	6.2	0.0	1,000	1	14.20
Phishing	MLP	LIME (full)	60	7.1	0.0	1,000	1	14.20
Phishing	MLP	Anchors	60	2,250	2,227	46,201	463	656.05
Phishing	MLP	DiCE	60	501.2	500.0	2,00	8	28.49

Dataset	Model	Explainer	n	Wall (ms)	CPU (ms)	Rows	Model calls	MFLOPs
						6		
Connect-4	Logistic regression	SHAP	60	36.4	31.2	25,001	2	6.35
Connect-4	Logistic regression	LIME (10)	60	8.6	15.6	1,000	1	0.25
Connect-4	Logistic regression	LIME (full)	60	10.9	15.6	1,000	1	0.25
Connect-4	Logistic regression	Anchors	60	1,223	1,203	8,401	85	2.13
Connect-4	Logistic regression	DiCE	60	1,418	1,406	2,004	6	0.51
Connect-4	Random forest	SHAP	60	49.5	46.9	0	0	—
Connect-4	Random forest	LIME (10)	60	25.7	31.2	1,000	1	—
Connect-4	Random forest	LIME (full)	60	28.7	31.2	1,000	1	—
Connect-4	Random forest	Anchors	60	7,198	7,164	36,151	362	—
Connect-4	Random forest	DiCE	60	1,482	1,469	2,004	6	—
Connect-4	XGBoost	SHAP	60	1.8	0.0	0	0	—
Connect-4	XGBoost	LIME (10)	60	12.9	15.6	1,000	1	—
Connect-4	XGBoost	LIME (full)	60	15.3	15.6	1,000	1	—
Connect-4	XGBoost	Anchors	60	3,643	3,617	33,701	338	—
Connect-4	XGBoost	DiCE	60	1,469	1,469	2,005	7	—
Connect-4	SVM (RBF)	SHAP	60	29,507	29,297	25,001	2	—
Connect-4	SVM (RBF)	LIME (10)	60	1,193	1,188	1,000	1	—
Connect-4	SVM (RBF)	LIME (full)	60	1,192	1,188	1,000	1	—

Dataset	Model	Explainer	n	Wall (ms)	CPU (ms)	Rows	Model calls	MFLOPs
Connect-4	SVM (RBF)	Anchors	60	27,862	27,453	21,351	214	—
Connect-4	SVM (RBF)	DiCE	60	3,788	3,766	2,004	6	—
Connect-4	MLP	SHAP	60	59.8	62.5	25,001	2	645.03
Connect-4	MLP	LIME (10)	60	9.8	15.6	1,000	1	25.80
Connect-4	MLP	LIME (full)	60	12.1	15.6	1,000	1	25.80
Connect-4	MLP	Anchors	60	3,978	3,953	33,351	334	860.46
Connect-4	MLP	DiCE	60	1,424	1,422	2,004	6	51.70

Anchors calls XGBoost through a prediction method the main timing pass did not count; its rows and calls on XGBoost come from a recount with the fixed counter (23 cells, same seeds; scripts/paper_b_xgb_anchors_rows.ps1), and its time from the main pass.

Matched tree ensembles (Adult, 100 trees each, three seeds, 20 instances per cell, same timing protocol): each ensemble at its own depth and at the other's. Median wall time per explanation:

Model	TreeSHAP (ms)	LIME, full (ms)	TreeSHAP / LIME
Random forest, depth 15	21.7	21.6	1.01
Random forest, depth 6	3.2	16.7	0.19
XGBoost, depth 15	7.4	15.9	0.46
XGBoost, depth 6	1.4	12.5	0.11

S10. Configuration sweep

Adult, five models, three seeds, one instance per stratum (12 per model), stability on each. LIME with the full vector. Degenerate explanations per configuration (pooled over models): share of all-zero explanations and mean number of all-zero vectors among the 15 perturbed explanations:

Configuration	All-zero	Zero among 15 perturbed
`kshap_ns1000_bg50`	0%	0.0
`kshap_ns100_bg50`	0%	0.0
`kshap_ns250_bg50`	0%	0.0

Configuration	All-zero	Zero among 15 perturbed
kshap_ns500_bg10	0%	0.0
kshap_ns500_bg100	0%	0.0
kshap_ns500_bg25	0%	0.0
kshap_ns500_bg50	0%	0.0
kshap_nsauto_bg50	0%	0.0
lime_kw0p75_ns1000	100%	15.0
lime_kw0p75_ns2000	100%	15.0
lime_kw0p75_ns250	100%	15.0
lime_kw0p75_ns500	100%	15.0
lime_kw0p75_ns5000	100%	15.0
lime_kw10p0_ns1000	0%	0.1
lime_kw10p0_ns2000	0%	0.1
lime_kw10p0_ns250	0%	0.1
lime_kw10p0_ns500	0%	0.1
lime_kw10p0_ns5000	0%	0.1
lime_kw15p0_ns1000	0%	0.0
lime_kw15p0_ns2000	0%	0.0
lime_kw15p0_ns250	0%	0.0
lime_kw15p0_ns500	0%	0.0
lime_kw15p0_ns5000	0%	0.0
lime_kw1p5_ns1000	100%	15.0
lime_kw1p5_ns2000	100%	15.0
lime_kw1p5_ns250	100%	15.0
lime_kw1p5_ns500	100%	15.0
lime_kw1p5_ns5000	100%	15.0
lime_kw3p0_ns1000	13%	15.0
lime_kw3p0_ns2000	13%	15.0
lime_kw3p0_ns250	13%	15.0
lime_kw3p0_ns500	13%	15.0
lime_kw3p0_ns5000	12%	15.0
lime_kw5p0_ns1000	3%	8.5
lime_kw5p0_ns2000	3%	8.0
lime_kw5p0_ns250	3%	10.1
lime_kw5p0_ns500	3%	9.7

Configuration	All-zero	Zero among 15 perturbed
lime_kw5p0_ns5000	0%	7.4
lime_kwdef_ns1000	0%	1.1
lime_kwdef_ns2000	0%	0.8
lime_kwdef_ns250	0%	1.3
lime_kwdef_ns500	0%	0.9
lime_kwdef_ns5000	0%	0.6
tshap_bg10	0%	0.0
tshap_bg100	0%	0.0
tshap_bg25	0%	0.0
tshap_bg50	0%	0.0

Mean stability, fidelity and faithfulness gap on the 12 instances valid under every remaining configuration (LIME widths 0.75 to 5 left out), median wall time per explanation from the sweep's timing pass (same protocol as S9), and membership of the stability-fidelity Pareto front (P):

Model	Configuration	Stability	Fidelity	Faithfulness gap	Wall (ms)	Front
Logistic regression	kshap_ns1000_bg50	0.90	0.96	0.31	50.9	P
Logistic regression	kshap_ns100_bg50	0.64	0.91	0.30	12.5	
Logistic regression	kshap_ns250_bg50	0.84	0.95	0.30	19.0	
Logistic regression	kshap_ns500_bg10	0.84	0.89	0.30	18.4	
Logistic regression	kshap_ns500_bg100	0.89	0.97	0.31	43.3	P
Logistic regression	kshap_ns500_bg25	0.88	0.95	0.30	22.5	
Logistic regression	kshap_ns500_bg50	0.89	0.97	0.30	29.1	
Logistic regression	kshap_nsauto_bg50	0.89	0.97	0.30	103.4	
Logistic regression	lime_kw10p0_ns1000	0.87	0.55	0.23	9.3	
Logistic regression	lime_kw10p0_ns2000	0.90	0.55	0.23	19.6	

Model	Configuration	Stability	Fidelity	Faithfulness gap	Wall (ms)	Front
Logistic regression	lime_kw10p0_ns250	0.69	0.51	0.24	4.2	
Logistic regression	lime_kw10p0_ns500	0.81	0.54	0.23	5.6	
Logistic regression	lime_kw10p0_ns5000	0.96	0.55	0.24	47.3	
Logistic regression	lime_kw15p0_ns1000	0.97	0.55	0.23	9.3	
Logistic regression	lime_kw15p0_ns2000	0.99	0.55	0.24	19.6	
Logistic regression	lime_kw15p0_ns250	0.84	0.51	0.24	4.1	
Logistic regression	lime_kw15p0_ns500	0.93	0.54	0.23	5.6	
Logistic regression	lime_kw15p0_ns5000	0.99	0.55	0.24	47.3	P
Logistic regression	lime_kwdef_ns1000	0.59	0.55	0.25	9.3	
Logistic regression	lime_kwdef_ns2000	0.73	0.55	0.22	19.6	
Logistic regression	lime_kwdef_ns250	0.43	0.51	0.24	4.3	
Logistic regression	lime_kwdef_ns500	0.55	0.54	0.23	5.5	
Logistic regression	lime_kwdef_ns5000	0.79	0.55	0.24	47.8	
Random forest	kshap_ns1000_bg50	0.98	0.89	0.37	360.3	
Random forest	kshap_ns100_bg50	0.79	0.88	0.37	51.4	
Random forest	kshap_ns250_bg50	0.96	0.90	0.37	109.7	
Random forest	kshap_ns500_bg10	0.97	0.82	0.37	56.1	
Random forest	kshap_ns500_bg100	0.98	0.89	0.38	352.3	
Random forest	kshap_ns500_bg25	0.97	0.88	0.37	112.7	

Model	Configuration	Stability	Fidelity	Faithfulness gap	Wall (ms)	Front
Random forest	kshap_ns500_bg50	0.97	0.89	0.37	189.2	
Random forest	kshap_nsauto_bg50	0.98	0.90	0.37	790.5	
Random forest	lime_kw10p0_ns1000	0.88	0.38	0.27	21.4	
Random forest	lime_kw10p0_ns2000	0.88	0.38	0.25	40.2	
Random forest	lime_kw10p0_ns250	0.60	0.31	0.24	11.0	
Random forest	lime_kw10p0_ns500	0.75	0.36	0.24	14.4	
Random forest	lime_kw10p0_ns5000	0.91	0.39	0.22	91.4	
Random forest	lime_kw15p0_ns1000	0.92	0.38	0.27	21.4	
Random forest	lime_kw15p0_ns2000	0.96	0.38	0.25	39.7	
Random forest	lime_kw15p0_ns250	0.68	0.31	0.24	11.2	
Random forest	lime_kw15p0_ns500	0.84	0.36	0.24	14.5	
Random forest	lime_kw15p0_ns5000	0.98	0.39	0.22	91.3	
Random forest	lime_kwdef_ns1000	0.52	0.38	0.25	21.3	
Random forest	lime_kwdef_ns2000	0.68	0.38	0.25	39.6	
Random forest	lime_kwdef_ns250	0.35	0.31	0.24	11.0	
Random forest	lime_kwdef_ns500	0.46	0.36	0.23	14.7	
Random forest	lime_kwdef_ns5000	0.72	0.39	0.22	92.2	
Random forest	tshap_bg10	0.97	0.82	0.36	8.7	
Random forest	tshap_bg100	0.99	0.90	0.38	47.3	P
Random forest	tshap_bg25	0.98	0.89	0.37	14.8	
Random forest	tshap_bg50	0.99	0.90	0.37	25.5	P
XGBoost	kshap_ns1000_bg50	0.62	0.72	0.44	198.1	

Model	Configuration	Stability	Fidelity	Faithfulness gap	Wall (ms)	Front
XGBoost	kshap_ns100_bg50	0.55	0.69	0.44	28.4	
XGBoost	kshap_ns250_bg50	0.61	0.71	0.44	57.1	
XGBoost	kshap_ns500_bg10	0.58	0.66	0.47	34.5	
XGBoost	kshap_ns500_bg100	0.63	0.74	0.45	189.5	P
XGBoost	kshap_ns500_bg25	0.62	0.70	0.44	61.1	
XGBoost	kshap_ns500_bg50	0.62	0.72	0.44	102.6	
XGBoost	kshap_nsauto_bg50	0.62	0.72	0.44	413.1	
XGBoost	lime_kw10p0_ns1000	0.84	0.58	0.43	12.7	
XGBoost	lime_kw10p0_ns2000	0.85	0.59	0.46	26.3	
XGBoost	lime_kw10p0_ns250	0.48	0.49	0.34	5.7	
XGBoost	lime_kw10p0_ns500	0.69	0.55	0.44	7.8	
XGBoost	lime_kw10p0_ns5000	0.95	0.60	0.46	63.4	P
XGBoost	lime_kw15p0_ns1000	0.89	0.58	0.43	12.7	
XGBoost	lime_kw15p0_ns2000	0.94	0.59	0.46	26.5	
XGBoost	lime_kw15p0_ns250	0.58	0.48	0.34	5.7	
XGBoost	lime_kw15p0_ns500	0.78	0.55	0.44	7.9	
XGBoost	lime_kw15p0_ns5000	0.98	0.60	0.46	62.6	P
XGBoost	lime_kwdef_ns1000	0.50	0.58	0.43	12.9	
XGBoost	lime_kwdef_ns2000	0.68	0.59	0.46	26.7	
XGBoost	lime_kwdef_ns250	0.32	0.50	0.35	5.7	
XGBoost	lime_kwdef_ns	0.41	0.56	0.44	7.9	

Model	Configuration	Stability	Fidelity	Faithfulness gap	Wall (ms)	Front
	500					
XGBoost	lime_kwdef_ns5000	0.74	0.60	0.46	63.3	P
XGBoost	tshap_bg10	0.61	0.65	0.47	0.4	
XGBoost	tshap_bg100	0.62	0.76	0.45	2.7	P
XGBoost	tshap_bg25	0.62	0.70	0.46	0.8	
XGBoost	tshap_bg50	0.62	0.73	0.46	1.4	
SVM (RBF)	kshap_ns1000_bg50	0.94	0.86	0.36	38,250	P
SVM (RBF)	kshap_ns100_bg50	0.77	0.81	0.33	3,838	
SVM (RBF)	kshap_ns250_bg50	0.91	0.85	0.34	9,572	
SVM (RBF)	kshap_ns500_bg10	0.93	0.82	0.36	3,826	
SVM (RBF)	kshap_ns500_bg100	0.94	0.86	0.36	38,257	
SVM (RBF)	kshap_ns500_bg25	0.93	0.84	0.36	9,571	
SVM (RBF)	kshap_ns500_bg50	0.93	0.86	0.35	19,231	
SVM (RBF)	kshap_nsauto_bg50	0.94	0.86	0.35	82,525	P
SVM (RBF)	lime_kw10p0_ns1000	0.80	0.41	0.20	769.7	
SVM (RBF)	lime_kw10p0_ns2000	0.91	0.43	0.23	1,551	
SVM (RBF)	lime_kw10p0_ns250	0.61	0.38	0.23	195.8	
SVM (RBF)	lime_kw10p0_ns500	0.74	0.41	0.21	388.0	
SVM (RBF)	lime_kw10p0_ns5000	0.90	0.43	0.21	3,862	
SVM (RBF)	lime_kw15p0_ns1000	0.96	0.41	0.20	774.5	
SVM (RBF)	lime_kw15p0_ns2000	0.97	0.43	0.23	1,553	
SVM (RBF)	lime_kw15p0_ns250	0.80	0.37	0.23	194.9	
SVM (RBF)	lime_kw15p0_n	0.90	0.41	0.20	389.0	

Model	Configuration	Stability	Fidelity	Faithfulness gap	Wall (ms)	Front
	s500					
SVM (RBF)	lime_kw15p0_ns5000	0.99	0.43	0.22	3,887	P
SVM (RBF)	lime_kwdef_ns1000	0.53	0.41	0.20	771.4	
SVM (RBF)	lime_kwdef_ns2000	0.66	0.43	0.23	1,551	
SVM (RBF)	lime_kwdef_ns250	0.39	0.38	0.20	196.7	
SVM (RBF)	lime_kwdef_ns500	0.50	0.41	0.21	385.5	
SVM (RBF)	lime_kwdef_ns5000	0.73	0.43	0.21	3,869	
MLP	kshap_ns1000_bg50	0.44	0.80	0.58	89.3	
MLP	kshap_ns100_bg50	0.25	0.68	0.43	16.7	
MLP	kshap_ns250_bg50	0.38	0.75	0.50	30.2	
MLP	kshap_ns500_bg10	0.43	0.73	0.41	22.5	
MLP	kshap_ns500_bg100	0.42	0.77	0.45	82.9	
MLP	kshap_ns500_bg25	0.43	0.78	0.51	33.8	
MLP	kshap_ns500_bg50	0.43	0.77	0.46	50.9	
MLP	kshap_nsauto_bg50	0.49	0.81	0.58	185.0	P
MLP	lime_kw10p0_ns1000	0.66	0.19	0.16	9.9	
MLP	lime_kw10p0_ns2000	0.75	0.22	0.16	21.1	
MLP	lime_kw10p0_ns250	0.36	0.11	0.08	4.5	
MLP	lime_kw10p0_ns500	0.51	0.13	0.11	5.9	
MLP	lime_kw10p0_ns5000	0.86	0.22	0.23	52.2	
MLP	lime_kw15p0_ns1000	0.78	0.19	0.17	9.9	

Model	Configuration	Stability	Fidelity	Faithfulness gap	Wall (ms)	Front
MLP	lime_kw15p0_ns2000	0.88	0.23	0.16	21.0	P
MLP	lime_kw15p0_ns250	0.39	0.11	0.08	4.4	
MLP	lime_kw15p0_ns500	0.61	0.13	0.11	5.8	
MLP	lime_kw15p0_ns5000	0.95	0.22	0.23	52.1	P
MLP	lime_kwdef_ns1000	0.51	0.22	0.25	9.8	
MLP	lime_kwdef_ns2000	0.59	0.24	0.16	21.0	P
MLP	lime_kwdef_ns250	0.21	0.18	0.09	4.2	
MLP	lime_kwdef_ns500	0.36	0.19	0.11	6.0	
MLP	lime_kwdef_ns5000	0.65	0.24	0.23	51.3	P

Configuration names: lime_kw<width>_ns<samples> (kwdef = 0.75 times the square root of the number of features), kshap_ns<coalitions>_bg<background rows> (nsauto = the library default, 2M + 2048), tshap_bg<background rows>. Wall time of every configuration, the degenerate LIME widths included, is in results/studies_timing/sweep_timing.csv.

S11. Measurement controls

Exclusion of instances with an all-zero LIME vector (any of the explanation and its 15 perturbed copies), Adult, mixed-model estimate of SHAP − X:

X	Endpoint	Instances	n	Estimate [95% CI]	p
LIME (full)	Stability	all	300	0.181 [−0.010, 0.372]	0.063
LIME (full)	Stability	without all-zero	134	0.129 [−0.066, 0.323]	0.196
LIME (full)	Fidelity	all	2,500	0.384 [0.225, 0.544]	<0.001
LIME (full)	Fidelity	without all-zero	2,326	0.384 [0.274, 0.494]	<0.001
LIME (full)	Faithfulness gap	all	2,500	0.131 [0.051, 0.212]	0.001
LIME (full)	Faithfulness gap	without all-zero	2,326	0.132 [0.026, 0.239]	0.014
LIME (10)	Stability	all	300	0.408 [0.247, 0.570]	<0.001
LIME (10)	Stability	without all-zero	155	0.387 [0.233, 0.541]	<0.001

X	Endpoint	Instances	n	Estimate [95% CI]	p
LIME (10)	Fidelity	all	2,500	0.314 [0.153, 0.476]	<0.001
LIME (10)	Fidelity	without all-zero	2,347	0.315 [0.201, 0.429]	<0.001
LIME (10)	Faithfulness gap	all	2,500	0.091 [−0.034, 0.217]	0.154
LIME (10)	Faithfulness gap	without all-zero	2,347	0.090 [0.033, 0.148]	0.002

Stability under the main protocol (Gaussian noise, sigma 0.1), without noise, and with noise that keeps one-hot features valid (a category resampled with probability 0.1), on the same Adult instances (three seeds, 12 stability instances per cell):

Model	Explainer	Main	No noise	Valid one-hot
Logistic regression	SHAP	0.87	0.98	0.82
Logistic regression	LIME (10)	0.40	0.99	0.87
Logistic regression	LIME (full)	0.60	0.97	0.95
Logistic regression	Anchors	0.64	0.77	0.55
Logistic regression	DiCE	0.38	0.37	0.37
Random forest	SHAP	0.97	1.00	0.86
Random forest	LIME (10)	0.40	0.98	0.86
Random forest	LIME (full)	0.59	0.93	0.92
Random forest	Anchors	0.64	0.77	0.57
Random forest	DiCE	0.49	0.51	0.41
XGBoost	SHAP	0.62	1.00	0.77
XGBoost	LIME (10)	0.23	0.98	0.85
XGBoost	LIME (full)	0.56	0.89	0.89
XGBoost	Anchors	0.56	0.73	0.49
XGBoost	DiCE	0.60	0.42	0.46
SVM (RBF)	SHAP	0.91	0.98	0.87
SVM (RBF)	LIME (10)	0.39	0.99	0.90
SVM (RBF)	LIME (full)	0.58	0.96	0.95
SVM (RBF)	Anchors	0.69	0.79	0.60
SVM (RBF)	DiCE	0.43	0.44	0.44
MLP	SHAP	0.34	0.92	0.67
MLP	LIME (10)	0.20	0.77	0.75
MLP	LIME (full)	0.46	0.78	0.78
MLP	Anchors	0.63	0.74	0.51

Model	Explainer	Main	No noise	Valid one-hot
MLP	DiCE	0.16	0.15	0.15

Stability of the main runs on Adult by measure: cosine (as in the paper), cosine without all-zero vectors, mean pairwise Spearman correlation of absolute attributions, and mean top-5 Jaccard overlap:

Model	Explainer	Cosine	Cosine, no zero vectors	Spearman	Top-5 Jaccard
Logistic regression	SHAP	0.86	0.86	0.56	0.63
Logistic regression	LIME (10)	0.41	0.48	0.42	0.32
Logistic regression	LIME (full)	0.59	0.66	0.25	0.46
Logistic regression	Anchors	0.63	0.63	0.62	0.50
Logistic regression	DiCE	0.35	0.35	0.23	0.17
Random forest	SHAP	0.98	0.98	1.00	0.92
Random forest	LIME (10)	0.37	0.42	0.35	0.26
Random forest	LIME (full)	0.56	0.64	0.19	0.42
Random forest	Anchors	0.66	0.66	0.64	0.54
Random forest	DiCE	0.50	0.50	0.33	0.25
XGBoost	SHAP	0.61	0.61	0.98	0.74
XGBoost	LIME (10)	0.24	0.26	0.26	0.16
XGBoost	LIME (full)	0.55	0.61	0.12	0.37
XGBoost	Anchors	0.57	0.57	0.54	0.45
XGBoost	DiCE	0.58	0.58	0.38	0.28
SVM (RBF)	SHAP	0.90	0.90	0.64	0.68
SVM (RBF)	LIME (10)	0.40	0.44	0.38	0.27
SVM (RBF)	LIME (full)	0.59	0.66	0.20	0.43
SVM (RBF)	Anchors	0.69	0.69	0.68	0.57
SVM (RBF)	DiCE	0.41	0.41	0.27	0.21
MLP	SHAP	0.32	0.32	0.21	0.18

Model	Explainer	Cosine	Cosine, no zero vectors	Spearman	Top-5 Jaccard
MLP	LIME (10)	0.19	0.20	0.25	0.14
MLP	LIME (full)	0.45	0.51	0.28	0.19
MLP	Anchors	0.63	0.63	0.58	0.53
MLP	DiCE	0.15	0.15	0.08	0.07

Faithfulness gap SHAP − X on Adult under four masking schemes of the top-5 features (zero, training mean, draw from the training marginal, group mean with one-hot groups and education / education-num masked together): mean of cell means, and cells with SHAP ahead:

Model	X	Zero	Mean	Marginal	Grouped
Logistic regression	Anchors	0.137 (5/5)	0.111 (5/5)	0.101 (5/5)	0.108 (5/5)
Logistic regression	DiCE	0.233 (5/5)	0.206 (5/5)	0.177 (5/5)	0.187 (5/5)
Logistic regression	LIME (10)	0.080 (5/5)	0.066 (5/5)	0.061 (5/5)	0.077 (5/5)
Logistic regression	LIME (full)	0.088 (5/5)	0.077 (5/5)	0.056 (5/5)	0.087 (5/5)
Random forest	Anchors	0.149 (5/5)	0.149 (5/5)	0.078 (5/5)	0.127 (5/5)
Random forest	DiCE	0.243 (5/5)	0.243 (5/5)	0.150 (5/5)	0.195 (5/5)
Random forest	LIME (10)	0.086 (5/5)	0.086 (5/5)	0.061 (5/5)	0.053 (5/5)
Random forest	LIME (full)	0.114 (5/5)	0.114 (5/5)	0.064 (5/5)	0.084 (5/5)
XGBoost	Anchors	0.161 (5/5)	0.161 (5/5)	0.082 (5/5)	0.157 (5/5)
XGBoost	DiCE	0.175 (5/5)	0.175 (5/5)	0.120 (5/5)	0.156 (5/5)
XGBoost	LIME (10)	0.005 (4/5)	0.005 (4/5)	0.039 (5/5)	0.003 (3/5)
XGBoost	LIME (full)	0.017 (5/5)	0.017 (5/5)	0.048 (5/5)	0.014 (5/5)
SVM (RBF)	Anchors	0.066 (5/5)	0.058 (5/5)	0.027 (5/5)	0.048 (5/5)
SVM (RBF)	DiCE	0.222 (5/5)	0.198 (5/5)	0.129 (5/5)	0.170 (5/5)
SVM (RBF)	LIME (10)	0.085 (5/5)	0.110 (5/5)	0.060 (5/5)	0.082 (5/5)
SVM (RBF)	LIME (full)	0.117 (5/5)	0.142 (5/5)	0.089 (5/5)	0.114 (5/5)
MLP	Anchors	0.193 (5/5)	0.204 (5/5)	0.140 (5/5)	0.149 (5/5)
MLP	DiCE	0.314 (5/5)	0.323 (5/5)	0.259 (5/5)	0.211 (5/5)
MLP	LIME (10)	0.189 (5/5)	0.190 (5/5)	0.161 (5/5)	0.108 (5/5)
MLP	LIME (full)	0.300 (5/5)	0.308 (5/5)	0.259 (5/5)	0.145 (5/5)

S12. Further datasets: per model

Mean (SD) over the three seeds, per dataset, model and explainer. Table 4 of the paper averages these over the five models.

Dataset	Model	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
Breast Cancer	Logistic regression	SHAP	0.88 (0.01)	0.80 (0.01)	0.16 (0.01)	0.33 (0.00)
Breast Cancer	Logistic regression	LIME (10)	0.89 (0.02)	0.74 (0.03)	0.08 (0.00)	0.33 (0.00)
Breast Cancer	Logistic regression	LIME (full)	0.99 (0.00)	0.53 (0.02)	0.05 (0.01)	1.00 (0.00)
Breast Cancer	Logistic regression	Anchors	0.47 (0.07)	0.35 (0.03)	0.08 (0.00)	0.07 (0.00)
Breast Cancer	Logistic regression	DiCE	0.27 (0.05)	0.07 (0.02)	0.04 (0.01)	0.15 (0.01)
Breast Cancer	Random forest	SHAP	0.84 (0.01)	0.74 (0.01)	0.26 (0.00)	0.98 (0.00)
Breast Cancer	Random forest	LIME (10)	0.88 (0.02)	0.67 (0.01)	0.24 (0.00)	0.33 (0.00)
Breast Cancer	Random forest	LIME (full)	0.99 (0.00)	0.63 (0.00)	0.23 (0.00)	0.99 (0.00)
Breast Cancer	Random forest	Anchors	0.25 (0.02)	0.25 (0.03)	0.12 (0.00)	0.07 (0.00)
Breast Cancer	Random forest	DiCE	0.24 (0.01)	0.27 (0.02)	0.12 (0.01)	0.24 (0.02)
Breast Cancer	XGBoost	SHAP	0.81 (0.01)	0.59 (0.02)	0.27 (0.02)	0.89 (0.00)
Breast Cancer	XGBoost	LIME (10)	0.87 (0.01)	0.51 (0.02)	0.22 (0.01)	0.33 (0.00)
Breast Cancer	XGBoost	LIME (full)	0.98 (0.00)	0.46 (0.02)	0.21 (0.01)	0.99 (0.00)
Breast Cancer	XGBoost	Anchors	0.40 (0.03)	0.19 (0.03)	0.08 (0.02)	0.07 (0.00)
Breast Cancer	XGBoost	DiCE	0.24 (0.02)	0.25 (0.08)	0.09 (0.01)	0.17 (0.01)
Breast Cancer	SVM (RBF)	SHAP	0.85 (0.02)	0.68 (0.02)	0.14 (0.01)	0.33 (0.00)
Breast Cancer	SVM (RBF)	LIME (10)	0.86 (0.03)	0.67 (0.04)	0.08 (0.01)	0.33 (0.00)
Breast Cancer	SVM (RBF)	LIME (full)	1.00 (0.00)	0.47 (0.02)	0.04 (0.00)	0.99 (0.01)
Breast	SVM (RBF)	Anchors	0.39	0.26	0.06 (0.01)	0.07

Dataset	Model	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
Cancer			(0.06)	(0.01)		(0.00)
Breast Cancer	SVM (RBF)	DiCE	0.16 (0.04)	0.04 (0.01)	0.03 (0.01)	0.07 (0.01)
Breast Cancer	MLP	SHAP	0.83 (0.01)	0.64 (0.02)	0.21 (0.06)	0.33 (0.00)
Breast Cancer	MLP	LIME (10)	0.85 (0.02)	0.56 (0.01)	0.09 (0.03)	0.33 (0.00)
Breast Cancer	MLP	LIME (full)	0.99 (0.00)	0.39 (0.01)	0.06 (0.01)	1.00 (0.00)
Breast Cancer	MLP	Anchors	0.41 (0.05)	0.23 (0.02)	0.10 (0.03)	0.07 (0.00)
Breast Cancer	MLP	DiCE	0.32 (0.01)	0.11 (0.02)	0.06 (0.03)	0.12 (0.01)
German Credit	Logistic regression	SHAP	0.84 (0.00)	0.89 (0.00)	0.28 (0.01)	0.16 (0.00)
German Credit	Logistic regression	LIME (10)	0.84 (0.02)	0.80 (0.01)	0.19 (0.00)	0.16 (0.00)
German Credit	Logistic regression	LIME (full)	0.99 (0.00)	0.69 (0.01)	0.16 (0.01)	1.00 (0.00)
German Credit	Logistic regression	Anchors	0.74 (0.02)	0.56 (0.01)	0.23 (0.00)	0.07 (0.00)
German Credit	Logistic regression	DiCE	0.37 (0.04)	0.14 (0.02)	0.07 (0.00)	0.04 (0.00)
German Credit	Random forest	SHAP	0.99 (0.00)	0.74 (0.01)	0.17 (0.00)	0.91 (0.00)
German Credit	Random forest	LIME (10)	0.92 (0.01)	0.41 (0.01)	0.14 (0.01)	0.16 (0.00)
German Credit	Random forest	LIME (full)	0.93 (0.00)	0.36 (0.01)	0.14 (0.01)	0.98 (0.00)
German Credit	Random forest	Anchors	0.69 (0.04)	0.31 (0.01)	0.08 (0.00)	0.18 (0.01)
German Credit	Random forest	DiCE	0.31 (0.03)	0.14 (0.02)	0.06 (0.01)	0.05 (0.00)
German Credit	XGBoost	SHAP	0.64 (0.02)	0.64 (0.01)	0.29 (0.01)	0.84 (0.00)
German Credit	XGBoost	LIME (10)	0.90 (0.02)	0.36 (0.01)	0.19 (0.01)	0.16 (0.00)

Dataset	Model	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
German Credit	XGBoost	LIME (full)	0.88 (0.00)	0.35 (0.01)	0.18 (0.02)	0.99 (0.00)
German Credit	XGBoost	Anchors	0.62 (0.09)	0.41 (0.03)	0.25 (0.02)	0.10 (0.00)
German Credit	XGBoost	DiCE	0.22 (0.02)	0.22 (0.01)	0.15 (0.02)	0.03 (0.00)
German Credit	SVM (RBF)	SHAP	0.89 (0.01)	0.81 (0.01)	0.26 (0.01)	0.16 (0.00)
German Credit	SVM (RBF)	LIME (10)	0.90 (0.02)	0.64 (0.00)	0.17 (0.00)	0.16 (0.00)
German Credit	SVM (RBF)	LIME (full)	0.98 (0.00)	0.57 (0.00)	0.16 (0.00)	1.00 (0.00)
German Credit	SVM (RBF)	Anchors	0.67 (0.05)	0.42 (0.01)	0.17 (0.00)	0.09 (0.00)
German Credit	SVM (RBF)	DiCE	0.38 (0.08)	0.23 (0.01)	0.09 (0.00)	0.03 (0.00)
German Credit	MLP	SHAP	0.78 (0.01)	0.79 (0.01)	0.53 (0.03)	0.16 (0.00)
German Credit	MLP	LIME (10)	0.70 (0.01)	0.40 (0.00)	0.26 (0.02)	0.16 (0.00)
German Credit	MLP	LIME (full)	0.92 (0.00)	0.24 (0.00)	0.21 (0.01)	1.00 (0.00)
German Credit	MLP	Anchors	0.66 (0.04)	0.44 (0.02)	0.35 (0.02)	0.11 (0.00)
German Credit	MLP	DiCE	0.22 (0.05)	0.16 (0.02)	0.14 (0.01)	0.03 (0.00)
Diabetes	Logistic regression	SHAP	0.98 (0.00)	0.93 (0.00)	0.27 (0.00)	1.00 (0.00)
Diabetes	Logistic regression	LIME (10)	1.00 (0.00)	0.68 (0.01)	0.27 (0.00)	1.00 (0.00)
Diabetes	Logistic regression	LIME (full)	1.00 (0.00)	0.68 (0.01)	0.27 (0.00)	1.00 (0.00)
Diabetes	Logistic regression	Anchors	0.89 (0.04)	0.54 (0.01)	0.28 (0.00)	0.29 (0.00)
Diabetes	Logistic regression	DiCE	0.36 (0.03)	0.36 (0.09)	0.21 (0.01)	0.19 (0.02)
Diabetes	Random forest	SHAP	0.94	0.71	0.26 (0.00)	0.99

Dataset	Model	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
			(0.01)	(0.00)		(0.00)
Diabetes	Random forest	LIME (10)	1.00 (0.00)	0.38 (0.02)	0.24 (0.00)	1.00 (0.00)
Diabetes	Random forest	LIME (full)	1.00 (0.00)	0.38 (0.02)	0.24 (0.00)	1.00 (0.00)
Diabetes	Random forest	Anchors	0.82 (0.04)	0.28 (0.03)	0.24 (0.00)	0.49 (0.01)
Diabetes	Random forest	DiCE	0.37 (0.04)	0.30 (0.06)	0.20 (0.01)	0.20 (0.01)
Diabetes	XGBoost	SHAP	0.86 (0.03)	0.75 (0.01)	0.38 (0.01)	1.00 (0.00)
Diabetes	XGBoost	LIME (10)	0.99 (0.00)	0.45 (0.02)	0.33 (0.01)	1.00 (0.00)
Diabetes	XGBoost	LIME (full)	0.99 (0.00)	0.45 (0.02)	0.33 (0.01)	1.00 (0.00)
Diabetes	XGBoost	Anchors	0.79 (0.01)	0.29 (0.00)	0.33 (0.00)	0.53 (0.00)
Diabetes	XGBoost	DiCE	0.32 (0.03)	0.33 (0.03)	0.31 (0.02)	0.20 (0.00)
Diabetes	SVM (RBF)	SHAP	0.98 (0.00)	0.81 (0.02)	0.27 (0.01)	1.00 (0.00)
Diabetes	SVM (RBF)	LIME (10)	1.00 (0.00)	0.48 (0.04)	0.26 (0.01)	1.00 (0.00)
Diabetes	SVM (RBF)	LIME (full)	1.00 (0.00)	0.48 (0.04)	0.26 (0.01)	1.00 (0.00)
Diabetes	SVM (RBF)	Anchors	0.78 (0.03)	0.37 (0.01)	0.27 (0.01)	0.38 (0.01)
Diabetes	SVM (RBF)	DiCE	0.28 (0.01)	0.14 (0.06)	0.17 (0.01)	0.20 (0.01)
Diabetes	MLP	SHAP	0.98 (0.00)	0.78 (0.00)	0.31 (0.01)	1.00 (0.00)
Diabetes	MLP	LIME (10)	1.00 (0.00)	0.46 (0.01)	0.30 (0.01)	1.00 (0.00)
Diabetes	MLP	LIME (full)	1.00 (0.00)	0.46 (0.01)	0.30 (0.01)	1.00 (0.00)
Diabetes	MLP	Anchors	0.81 (0.03)	0.35 (0.04)	0.32 (0.01)	0.35 (0.00)

Dataset	Model	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
Diabetes	MLP	DiCE	0.28 (0.02)	0.21 (0.02)	0.21 (0.01)	0.19 (0.01)
Spambase	Logistic regression	SHAP	0.78 (0.06)	0.75 (0.01)	0.36 (0.00)	0.18 (0.00)
Spambase	Logistic regression	LIME (10)	0.80 (0.02)	0.59 (0.01)	0.22 (0.01)	0.17 (0.00)
Spambase	Logistic regression	LIME (full)	0.98 (0.00)	0.52 (0.02)	0.21 (0.01)	0.99 (0.01)
Spambase	Logistic regression	Anchors	0.31 (0.07)	0.35 (0.01)	0.21 (0.01)	0.06 (0.00)
Spambase	Logistic regression	DiCE	0.06 (0.01)	0.00 (0.02)	0.09 (0.01)	0.03 (0.00)
Spambase	Random forest	SHAP	0.60 (0.02)	0.51 (0.02)	0.35 (0.02)	0.90 (0.00)
Spambase	Random forest	LIME (10)	0.83 (0.04)	0.59 (0.00)	0.24 (0.01)	0.17 (0.00)
Spambase	Random forest	LIME (full)	0.93 (0.04)	0.61 (0.00)	0.23 (0.01)	0.97 (0.00)
Spambase	Random forest	Anchors	0.29 (0.03)	0.11 (0.02)	0.21 (0.01)	0.05 (0.00)
Spambase	Random forest	DiCE	0.17 (0.03)	0.31 (0.01)	0.30 (0.00)	0.06 (0.00)
Spambase	XGBoost	SHAP	0.45 (0.03)	0.53 (0.01)	0.60 (0.01)	0.80 (0.00)
Spambase	XGBoost	LIME (10)	0.75 (0.05)	0.47 (0.00)	0.39 (0.02)	0.17 (0.00)
Spambase	XGBoost	LIME (full)	0.88 (0.04)	0.46 (0.00)	0.39 (0.02)	0.97 (0.01)
Spambase	XGBoost	Anchors	0.24 (0.01)	0.10 (0.01)	0.27 (0.02)	0.05 (0.00)
Spambase	XGBoost	DiCE	0.18 (0.01)	0.26 (0.02)	0.32 (0.01)	0.04 (0.00)
Spambase	SVM (RBF)	SHAP	0.90 (0.02)	0.80 (0.01)	0.36 (0.03)	0.18 (0.00)
Spambase	SVM (RBF)	LIME (10)	0.81 (0.01)	0.55 (0.00)	0.12 (0.01)	0.17 (0.00)
Spambas	SVM (RBF)	LIME	0.99	0.44	0.11 (0.01)	0.98

Dataset	Model	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
e		(full)	(0.00)	(0.00)		(0.00)
Spambase	SVM (RBF)	Anchors	0.34 (0.04)	0.39 (0.01)	0.21 (0.01)	0.06 (0.00)
Spambase	SVM (RBF)	DiCE	0.10 (0.05)	0.07 (0.00)	0.07 (0.01)	0.03 (0.00)
Spambase	MLP	SHAP	0.72 (0.03)	0.62 (0.03)	0.62 (0.05)	0.18 (0.00)
Spambase	MLP	LIME (10)	0.71 (0.02)	0.33 (0.02)	0.28 (0.03)	0.17 (0.00)
Spambase	MLP	LIME (full)	0.95 (0.00)	0.26 (0.03)	0.28 (0.03)	0.99 (0.01)
Spambase	MLP	Anchors	0.32 (0.04)	0.26 (0.01)	0.32 (0.02)	0.07 (0.00)
Spambase	MLP	DiCE	0.05 (0.01)	0.01 (0.02)	0.13 (0.01)	0.03 (0.00)
Phishing	Logistic regression	SHAP	0.83 (0.02)	0.87 (0.01)	0.41 (0.03)	0.15 (0.00)
Phishing	Logistic regression	LIME (10)	0.87 (0.00)	0.80 (0.00)	0.25 (0.02)	0.15 (0.00)
Phishing	Logistic regression	LIME (full)	0.99 (0.00)	0.78 (0.00)	0.24 (0.01)	1.00 (0.00)
Phishing	Logistic regression	Anchors	0.94 (0.01)	0.68 (0.01)	0.36 (0.01)	0.04 (0.00)
Phishing	Logistic regression	DiCE	0.25 (0.05)	0.29 (0.03)	0.14 (0.01)	0.05 (0.00)
Phishing	Random forest	SHAP	1.00 (0.00)	0.54 (0.01)	0.35 (0.00)	0.87 (0.00)
Phishing	Random forest	LIME (10)	0.92 (0.00)	0.30 (0.01)	0.25 (0.01)	0.15 (0.00)
Phishing	Random forest	LIME (full)	0.93 (0.00)	0.26 (0.01)	0.23 (0.00)	0.99 (0.00)
Phishing	Random forest	Anchors	0.86 (0.04)	0.38 (0.03)	0.27 (0.00)	0.06 (0.00)
Phishing	Random forest	DiCE	0.28 (0.03)	0.12 (0.02)	0.12 (0.01)	0.04 (0.00)
Phishing	XGBoost	SHAP	0.41 (0.03)	0.56 (0.01)	0.62 (0.01)	0.59 (0.00)

Dataset	Model	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
Phishing	XGBoost	LIME (10)	0.90 (0.01)	0.52 (0.01)	0.49 (0.00)	0.15 (0.00)
Phishing	XGBoost	LIME (full)	0.90 (0.00)	0.51 (0.01)	0.49 (0.00)	0.99 (0.00)
Phishing	XGBoost	Anchors	0.89 (0.00)	0.33 (0.02)	0.31 (0.01)	0.05 (0.00)
Phishing	XGBoost	DiCE	0.20 (0.04)	0.37 (0.06)	0.47 (0.03)	0.03 (0.00)
Phishing	SVM (RBF)	SHAP	0.86 (0.02)	0.87 (0.01)	0.44 (0.01)	0.15 (0.00)
Phishing	SVM (RBF)	LIME (10)	0.90 (0.00)	0.75 (0.01)	0.26 (0.04)	0.15 (0.00)
Phishing	SVM (RBF)	LIME (full)	0.99 (0.00)	0.72 (0.01)	0.23 (0.03)	1.00 (0.00)
Phishing	SVM (RBF)	Anchors	0.96 (0.04)	0.70 (0.02)	0.36 (0.02)	0.04 (0.00)
Phishing	SVM (RBF)	DiCE	0.31 (0.04)	0.36 (0.03)	0.13 (0.02)	0.07 (0.00)
Phishing	MLP	SHAP	0.65 (0.05)	0.65 (0.02)	0.48 (0.01)	0.15 (0.00)
Phishing	MLP	LIME (10)	0.86 (0.01)	0.47 (0.01)	0.19 (0.01)	0.15 (0.00)
Phishing	MLP	LIME (full)	0.96 (0.00)	0.45 (0.01)	0.19 (0.01)	1.00 (0.00)
Phishing	MLP	Anchors	0.81 (0.04)	0.46 (0.02)	0.32 (0.03)	0.08 (0.00)
Phishing	MLP	DiCE	0.20 (0.04)	0.17 (0.02)	0.16 (0.02)	0.04 (0.00)
Connect-4	Logistic regression	SHAP	0.73 (0.05)	0.88 (0.00)	0.28 (0.01)	0.08 (0.00)
Connect-4	Logistic regression	LIME (10)	0.61 (0.04)	0.68 (0.00)	0.20 (0.00)	0.08 (0.00)
Connect-4	Logistic regression	LIME (full)	0.98 (0.00)	0.55 (0.01)	0.14 (0.01)	0.99 (0.01)
Connect-4	Logistic regression	Anchors	0.74 (0.03)	0.62 (0.00)	0.39 (0.01)	0.04 (0.00)
Connect-	Logistic	DiCE	0.06	0.06	0.05 (0.01)	0.02

Dataset	Model	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
4	regression		(0.01)	(0.01)		(0.00)
Connect-4	Random forest	SHAP	1.00 (0.00)	0.80 (0.01)	0.20 (0.01)	0.67 (0.00)
Connect-4	Random forest	LIME (10)	0.72 (0.03)	0.47 (0.01)	0.16 (0.01)	0.08 (0.00)
Connect-4	Random forest	LIME (full)	0.82 (0.01)	0.32 (0.02)	0.14 (0.01)	0.98 (0.01)
Connect-4	Random forest	Anchors	0.68 (0.05)	0.40 (0.01)	0.15 (0.01)	0.07 (0.01)
Connect-4	Random forest	DiCE	0.19 (0.05)	0.10 (0.01)	0.05 (0.00)	0.02 (0.00)
Connect-4	XGBoost	SHAP	0.48 (0.04)	0.77 (0.01)	0.25 (0.03)	0.64 (0.00)
Connect-4	XGBoost	LIME (10)	0.61 (0.02)	0.32 (0.01)	0.16 (0.01)	0.08 (0.00)
Connect-4	XGBoost	LIME (full)	0.65 (0.03)	0.23 (0.01)	0.15 (0.02)	0.97 (0.01)
Connect-4	XGBoost	Anchors	0.72 (0.02)	0.41 (0.05)	0.28 (0.05)	0.06 (0.01)
Connect-4	XGBoost	DiCE	0.08 (0.02)	0.18 (0.03)	0.13 (0.03)	0.02 (0.00)
Connect-4	SVM (RBF)	SHAP	0.79 (0.01)	0.85 (0.00)	0.40 (0.01)	0.08 (0.00)
Connect-4	SVM (RBF)	LIME (10)	0.67 (0.01)	0.64 (0.01)	0.25 (0.01)	0.08 (0.00)
Connect-4	SVM (RBF)	LIME (full)	0.96 (0.00)	0.49 (0.01)	0.18 (0.03)	0.99 (0.00)
Connect-4	SVM (RBF)	Anchors	0.77 (0.05)	0.57 (0.01)	0.47 (0.03)	0.05 (0.00)
Connect-4	SVM (RBF)	DiCE	0.13 (0.02)	0.09 (0.01)	0.08 (0.01)	0.02 (0.00)
Connect-4	MLP	SHAP	0.48 (0.05)	0.69 (0.03)	0.71 (0.03)	0.08 (0.00)
Connect-4	MLP	LIME (10)	0.45 (0.02)	0.36 (0.01)	0.34 (0.01)	0.08 (0.00)
Connect-4	MLP	LIME (full)	0.83 (0.00)	0.22 (0.01)	0.27 (0.05)	1.00 (0.00)

Dataset	Model	Explainer	Stability	Fidelity	Faithfulness gap	Sparsity
Connect-4	MLP	Anchors	0.67 (0.04)	0.44 (0.02)	0.53 (0.03)	0.08 (0.00)
Connect-4	MLP	DiCE	0.05 (0.02)	0.03 (0.01)	0.08 (0.01)	0.01 (0.00)

S13. How to regenerate every table

1. Install the frozen environment; prepare the datasets and check the model hashes of S6 (`paper_b_run.py prepare`, then `paper_b_run.py preflight`). The prepared data reproduce the data hashes of S6 exactly.
2. Quality pass: `paper_b_run.py launch` (resumable; one row per instance).
3. Cost pass: `paper_b_run.py launch-timing` on an idle host.
4. Analysis: `paper_b_analyze.py --root <results> --out <analysis>` writes every value of Tables 2–4 of the paper and of S7–S12 (primary_effects.csv, cells.csv, cost.csv, quality_checks.csv, cross_dataset.csv, friedman.csv).
5. This file: `build_supplement.py --analysis <analysis>` derives the instance counts, overlaps, distributions, Figure S1 and the exclusion check from the per-instance rows.